Minimax Optimality in Contextual Dynamic Pricing with General Valuation Models

Xueping Gong, Wei You, Jiheng Zhang
The Hong Kong University of Science and Technology, Clear Water Bay, Hong Kong S.A.R., China.
xgongah@connect.ust.hk, weiyou@ust.hk, jiheng@ust.hk

We study the contextual dynamic pricing problem, where a decision maker selects personalized prices to maximize revenue based on observable contexts. The decision maker receives binary purchase feedback, which reflects whether customers' valuations exceed the posted prices. Each customer's valuation is determined by a latent function of the context, further corrupted by a market noise that follows an unknown distribution. We propose a minimax optimal algorithm under minimal assumptions, specifically, requiring only a Lipschitz-continuous noise distribution and bounded customer valuations. To address the unknown noise distribution, our method first discretizes its domain and constructs a finite set of candidate prices. We then employ a layered data partitioning technique on these prices, which yields tighter confidence bounds compared to those derived from the elliptical potential lemma. This technique also addresses the unknown bias stemming from estimation errors in the valuation model, hence eliminating the need to know the Lipschitz constant of the noise distribution. Moreover, our framework naturally extends to general valuation function spaces beyond the traditional linear model by incorporating offline regression oracles. Our regret analysis depends solely on the estimation error of these oracles, typically measured by the statistical complexity $\rho_{\mathcal{V}}(\delta)$ of the underlying function space $\mathcal{V}$. These techniques lead to a regret upper bound of $\tilde{\mathcal{O}}\left(\rho_{\mathcal{V}}(\delta)^{\frac{1}{3}}T^{\frac{2}{3}}\right)$, which matches the lower bound up to logarithmic factors, thus establishing the minimax optimality of our approach. Finally, numerical experiments validate our theoretical findings and demonstrate clear improvements over benchmark approaches.

Key words: Contextual dynamic pricing, layered data partitioning, minimax regret, general valuation models.

1. Introduction

The dynamic pricing problem, which involves setting real-time prices for products or services, has received significant attention due to its practical applications and direct impact on revenue maximization in industries such as entertainment, e-commerce (Lei et al. 2018), and transportation (Saharan et al. 2020). For an extensive review of the dynamic pricing literature, we recommend Den Boer (2015). Recent research has increasingly focused on feature-based dynamic pricing models that leverage observable contexts to understand market value and design effective pricing strategies (Fan et al. 2024, Cesa-Bianchi et al. 2019, Chen et al. 2024). These models capture product heterogeneity and enable personalized pricing.

At each decision epoch, the seller observes covariates that represent relevant product features and customer characteristics. These covariates determine the customer's valuation through an unknown valuation function and market noise. The customer purchases the product if their valuation exceeds the posted price. In standard settings where only binary purchase decisions (0-1 signals) are observed, the seller receives *censored* feedback about the customer's latent valuation. The goal is to set prices adaptively to maximize revenue while simultaneously learning the unknown valuation function and noise distribution.

Designing low-regret policies in this setting is challenging. The demand curve shifts continuously as covariates change, and its shape may not adhere to any specific parametric form and is often obscured by market noise. Consequently, solving the contextual dynamic pricing problem requires precise estimation of the demand function over a broad range of prices and contexts. This is in stark contrast to traditional non-contextual pricing or nonparametric bandit problems, where estimation around a single maximum suffices. Without a pricing policy that is carefully tailored to these complexities, regret can be significantly higher.

Despite extensive research on the contextual dynamic pricing problem, a gap remains in developing policies that are provably optimal while relying on minimal assumptions. Existing approaches often require strong assumptions about the smoothness of the valuation function or the noise distribution, which may not hold in practice. For instance, Javanmard and Nazerzadeh (2019) assume a known noise distribution, Tullii et al. (2024) require that the Lipschitz constant of the noise distribution is known, and Fan et al. (2024) require the noise distribution to be m-th differentiable. In this work, we overcome these limitations by developing methods that achieve improved regret bounds while imposing minimal assumptions.

To tackle the problem, we adopt an explore-then-UCB strategy within an epoch-based learning framework. Initially, a pure exploration phase gathers samples to estimate the true valuation function. Next, the algorithm enters a UCB phase, where we discretize the unknown distribution's domain into equal-length intervals. This discretization induces a linear bandit structure that enables robust upper confidence bounds accounting for estimation errors and biases. Our improved regret bound is achieved by exploiting this structure with a tighter concentration inequality to control regret within each episode. This concentration inequality requires delicate data independence, which we ensure through a layered data partitioning technique. Notably, the adaptive nature of this technique eliminates the need for prior knowledge of the Lipschitz constant of the noise distribution, enhancing flexibility and practicality. Finally, by carefully selecting the number of discretization intervals, we balance the learning and discretization regrets, ultimately achieving minimax optimality.

1.1. Our contributions

A novel algorithm with minimal assumptions. We propose a minimax optimal algorithm for contextual dynamic pricing that requires under minimal assumptions, requiring only bounded valuation and Lipschitz continuous noise, which are common in the literature (Javanmard and Nazerzadeh 2019, Chen and Gallego 2021, Choi et al. 2023, Fan et al. 2024, Luo et al. 2024). In contrast to existing methods that depend on stronger assumptions such as known noise distributions, second-order smoothness, or uniqueness of the optimal price, our algorithm requires none of these, representing a significant improvement. Our algorithm achieves a regret upper bound of $\tilde{\mathcal{O}}\left(\rho_{\mathcal{V}}(\delta)^{\frac{1}{3}}T^{\frac{2}{3}}\right)$, where $\rho_{\mathcal{V}}(\delta)$ captures the statistical complexity of the valuation function space. This bound matches the lower bound up to logarithmic factors, closing the gap in the literature. As a by-product, we extend existing lower bounds for dynamic pricing problems to cover smoother distributions beyond mere Lipschitz continuity (see Theorem 4.2).

Improved regret bounds for linear valuation models. When applied to linear valuation models with d_0-dimensional covariates, our results yield $\tilde{\mathcal{O}}(d_0^{\frac{1}{3}}T^{\frac{2}{3}})$ regret as $\rho_{\mathcal{V}}(\delta) = \mathcal{O}(d_0 \ln(1/\delta))$. This improves significantly upon prior works, such as $\tilde{\mathcal{O}}(d_0^2 T^{\frac{2}{3}})$ (with additional second-order smoothness assumption) and $\tilde{\mathcal{O}}(d_0 T^{\frac{3}{4}})$ (without it) in Luo et al. (2024). Moreover, compared to Fan et al. (2024), who obtain $\tilde{\mathcal{O}}(d_0^{\frac{5}{7}}T^{\frac{5}{7}})$ regret upper bound, our approach achieves a lower order in T while relaxing assumptions. In Table 1, we provide a comprehensive comparison with existing methods under linear valuation models, highlighting the improvements achieved by our approach in terms of reduced regret bounds and relaxed assumptions.

Table 1 Comparison of existing methods for linear valuation models with d_0 dimensional feature.

Method	**Regret***	**Additional Assumptions**†
Luo et al. (2022, 2024)	$d_0 T^{\frac{3}{4}}$	
	$d_0^2 T^{\frac{2}{3}}$	Second-order smoothness (Assumption 5.2)
	$d_0^2 T^{\frac{2}{3}} \vee d_0^{\frac{1}{2}} T^{1-\frac{\alpha}{2}}$	Availability of a classification oracle (Assumption 5.3)
Fan et al. (2024)	$(d_0 T)^{\frac{3}{4}}$	Uniqueness of the optimal price
	$(d_0 T)^{\frac{2m+1}{4m-1}}$	m-th differentiable, uniqueness of the optimal price
Tullii et al. (2024)	$(d_0 T)^{\frac{2}{3}}$	Known Lipschitz constant L in Assumption 2.2
Ours (Corollary 5.1)	$d_0^{\frac{1}{3}} T^{\frac{2}{3}}$	

Note:

* The regrets listed above omit constant factors and poly$(\log T)$ terms.

† All methods require Lipschitz continuity (Assumption 2.2) and boundedness of the valuation function (Assumption 2.1), which are therefore omitted in the table.

Generalization beyond linear models. Our method is flexible as it naturally extends beyond linear models by considering general function spaces and leveraging offline regression oracles. This enables us to handle a broad range of models, including sparse linear models and advanced frameworks such as reproducing kernel Hilbert spaces (RKHS) for the valuation function space. In contrast, many existing methods assume that the valuation function is linear (Luo et al. 2022, Fan et al. 2024, Javanmard and Nazerzadeh 2019), which limits their applicability to more complex settings. For example, Xu and Wang (2022) perform component-wise discretization of the parameter space, which is intrinsically tied to linearity assumption, and hence it is unclear how it may be extended to non-linear valuation structures. Cohen et al. (2020) employ ellipsoid-based shallow cuts to refine parameter uncertainty, so their geometric approach fundamentally relies on linearity. Recent work by Chen et al. (2024) relaxes linearity by proposing a nonparametric nearest-neighbor estimator for general valuation functions, which is a special offline regression oracle.

Improved regret bounds under additional information. Our framework also allows the incorporation of additional information to further reduce regret upper bounds. In particular, we study three special scenarios. In scenario (i), where the noise distribution function is known, our results are comparable to those of Javanmard and Nazerzadeh (2019) (which focus on linear models) under identical assumptions while our method extends to general valuation functions. In scenario (ii), where full valuations (beyond purchase decisions) are observable, we demonstrate that imposing an additional second-order smoothness assumption achieves a tighter regret rate of $\tilde{\mathcal{O}}(T^{\frac{3}{5}})$. Finally, in scenario (iii), under censored feedback with access to a classification oracle (Assumption 5.3), our method improves upon the $\tilde{\mathcal{O}}(T^{\frac{2}{3}\vee(1-\frac{\alpha}{2})})$ upper bounds in Luo et al. (2024), attaining $\tilde{\mathcal{O}}(T^{\frac{3}{5}\vee(1-\frac{\alpha}{2})})$, where α is the order of the estimation error in the classification oracle.

Layered data partitioning technique for dynamic pricing. To the best of our knowledge, we are the first to adapt the layered data partitioning technique to dynamic pricing. This technique partitions data into temporally decoupled layers to ensure statistical independence, enabling tight confidence bounds via Azuma's inequality. Therefore, our algorithm achieves $\tilde{\mathcal{O}}(d_0^{\frac{1}{3}}T^{\frac{2}{3}})$ regret, improving the prior state-of-the-art $\tilde{\mathcal{O}}(d_0T^{\frac{3}{4}})$. Moreover, this approach requires no prior knowledge of problem-specific parameters (e.g., the Lipschitz constant, compared with Tullii et al. 2024), which are rarely known in practice.

1.2. Related work

Dynamic pricing. Dynamic pricing has been an active area of research, driven by advancements in data technology and the increasing availability of customer information. Initial research focused

on non contextual dynamic pricing (Besbes and Zeevi 2015, Cesa-Bianchi et al. 2019). For example, Wang et al. (2021) employed the UCB approach with local-bin approximations, achieving an $\tilde{\mathcal{O}}(T^{\frac{m+1}{2m+1}})$ regret for m-th smooth demand functions and establishing a matching lower bound. However, these approaches do not incorporate covariates into pricing policies, making our model fundamentally different.

In the domain of dynamic pricing with covariates, the linear customer valuation model has been widely adopted. Javanmard and Nazerzadeh (2019) studied this model assuming a *known* and *log-concave* noise distribution. In contrast, Golrezaei et al. (2019) considered an ambiguity set for the noise distribution, achieving a regret of $\tilde{\mathcal{O}}(T^{\frac{2}{3}})$ compared to a robust benchmark, although their approach struggles when the ambiguity set encompasses an infinite number of distributions. Our model, however, addresses the general case of an unknown noise distribution and establishes regret bounds by comparing against the true optimal policy instead of a robust benchmark. Chen et al. (2024) explored nonparametric aspects of the unknown demand function using adaptive binning of the covariate space to achieve a regret of $\tilde{\mathcal{O}}(T^{\frac{d_0+2}{d_0+4}})$. Notably, our method outperforms theirs when $d_0 \geq 2$. Under the Cox proportional hazards (PH) model, Choi et al. (2023) introduced the CoxCP algorithm, which achieved a regret of $\tilde{\mathcal{O}}(T^{\frac{2}{3}})$. However, their approach relies on the separability of the unknown linear structure and noise distribution in the PH model, making it unsuitable for our setting where these components are entangled. Recent work by Xu and Wang (2022) proposed an adaptive pricing policy with $\tilde{\mathcal{O}}(T^{\frac{3}{4}})$ regret for adversarial contexts and bounded noise distributions. Our method improves this to $\tilde{\mathcal{O}}(T^{\frac{2}{3}})$ under Lipschitz noise distributions, while also maintaining computational efficiency compared to the exponential computations required by the EXP4-based policy in Xu and Wang (2022).

Other related studies (Luo et al. 2022, 2024, Fan et al. 2024) share similarities with our work in terms of settings, but differ in their assumptions about the smoothness of the noise distribution. For example, Luo et al. (2022) proposed an epoch-based algorithm with regret bounds of $\tilde{\mathcal{O}}(d_0^2 T^{\frac{2}{3}})$ and $\tilde{\mathcal{O}}(d_0 T^{\frac{3}{4}})$ under different smoothness assumptions. Fan et al. (2024) considered m-th differentiable distributions and achieved a regret of $\tilde{\mathcal{O}}((d_0 T)^{\frac{2m+1}{4m-1}})$ using the Nadaraya-Watson kernel regression estimator. In comparison, our method achieves $\tilde{\mathcal{O}}(d_0^{\frac{1}{3}} T^{\frac{2}{3}})$ regrets when applied to linear valuation models, demonstrating minimax optimality while relying on minimal assumptions.

Contextual bandits. Our pricing policy is closely related to bandit algorithms (Lattimore and Szepesvári 2020, Foster and Rakhlin 2020, Abbasi-Yadkori et al. 2011, Takemura et al. 2021, Auer 2002) that balance exploration and exploitation in decision-making. In particular, our approach connects to misspecified linear bandits (Auer 2002, Takemura et al. 2021). Unlike traditional bandit

algorithms, dynamic pricing must account for both the variance of estimation and the bias arising from parameter perturbations. We show that the unique structure of the dynamic pricing problem, when cast as a misspecified linear bandit, permits a more precise concentration bound, leading to improved regret bounds compared to the naive application of standard misspecified linear bandit algorithms. This improvement stems from the fact that the number of candidate prices is finite in each round. Our results highlight the importance of leveraging the distinctive structure of the pricing context to achieve optimal performance.

1.3. Notation and organization

Throughout the paper, we use the following notations. For any positive integer n, we denote the list $\{1, 2, \cdots, n\}$ as $[n]$. The cardinality of a set A is denoted by $|A|$. We use $\mathbf{I}_{\{E\}}$ to represent the indicator function of an event E. We denote by $\|\cdot\|_p$, for $1 \le p \le \infty$, the ℓ_p norm. Throughout the analysis, the notation $\tilde{\mathcal{O}}$ hides dependence on absolute constants and logarithmic terms.

In Section 2, we introduce the contextual dynamic pricing problem and present key assumptions of our approach. Section 3 presents the details of our algorithmic framework, with discussions of essential techniques and methodologies. Section 4 provides a theoretical analysis of the regret bounds for the proposed algorithm. Building on this framework, we discuss several extensions of our method in Section 5. In Section 6, we present numerical experiments comparing our methods with existing approaches. While proof sketches are integrated throughout the paper, comprehensive proofs are deferred to the Appendix.

2. Problem Formulation

In the contextual dynamic pricing setup, a potential customer arrives on the platform in each round $t \in [T]$, and the seller observes a covariate $x_t \in \mathcal{X} \subset \mathbb{R}^{d_0}$ that represents relevant product features and customer characteristics. Once x_t is observed, the customer's valuation for the product is $v_t = v^*(x_t) + \epsilon_t$, where $v^*(x_t)$ is an unknown valuation function and ϵ_t is the market noise drawn from an unknown distribution.

If the valuation v_t exceeds the posted price p_t, a sale occurs, and the seller earns revenue p_t. Otherwise, if $v_t < p_t$, no sale is made, and the revenue is zero. We denote the sale outcome by $y_t = \mathbf{I}_{\{v_t \ge p_t\}}$, so that y_t follows a Bernoulli distribution with parameter $1 - F(p_t - v^*(x_t))$. The revenue at time t is then $r_t = p_t y_t$. Thus, the triplet (x_t, p_t, y_t) summarizes the key information of the pricing process at round t.

Given the covariate x_t, setting a price p yields an expected reward of

$$\mathsf{Rev}_t(p) = p(1 - F(p - v^*(x_t))).$$

The optimal price maximizes the expected reward, i.e.,

$$p_t^* \in \arg\max_{p \geq 0} \mathsf{Rev}_t(p).$$

At each time t, the regret is defined as the difference between the revenue from the optimal price p_t^* and the revenue from the chosen price p_t. Over a horizon of T rounds, the cumulative regret is

$$\mathrm{Reg}(T) = \sum_{t=1}^{T} (\mathsf{Rev}_t(p_t^*) - \mathsf{Rev}_t(p_t)) = \sum_{t=1}^{T} \left[p_t^*(1 - F(p_t^* - v^*(x_t))) - p_t(1 - F(p_t - v^*(x_t))) \right].$$

We then consider the expected cumulative regret $\mathbb{E}[\mathrm{Reg}(T)]$, where the expectation accounts for randomness in both the covariates and the pricing policy. The primary objective in contextual dynamic pricing is to choose p_t for each covariate x_t at time t, leveraging historical data $\{(x_s, p_s, y_s), s \in [t-1]\}$, so as to minimize $\mathbb{E}[\mathrm{Reg}(T)]$.

We begin by outlining the assumptions made in this paper. We assume that the valuation function v^* belongs to a known function space $\mathcal{V}$, and that the noise terms $\{\epsilon_t\}_{t \in [T]}$ are independent and identically distributed according to an unknown zero-mean distribution F.

Assumption 2.1. (Bounded valuation) There exist positive, finite constants B_ϵ and B such that $|\epsilon_t| \leq B_\epsilon$ for all t, and $v^*(x) \in [B_\epsilon, B - B_\epsilon]$ for all $x \in \mathcal{X}$.

Assumption 2.1 imposes a known upper bound on the customers' valuations, which is a natural assumption for real-world products. Since $v^*(x)$ is bounded above by $B - B_\epsilon$, there is also a known universal upper bound B for the optimal price.

Next, we introduce the Lipschitz condition in Assumption 2.2, an assumption that has been widely studied in the literature (Luo et al. 2024, 2022, Chen et al. 2024). Our work also includes the case that F has a bounded derivative (Chen et al. 2024, Fan et al. 2024, Javanmard and Nazerzadeh 2019). Unlike several prior works (Luo et al. 2022, Fan et al. 2024, Javanmard and Nazerzadeh 2019), we do not assume, either explicitly or implicitly, that the optimal price is unique. Additionally, we do not make any concavity assumptions on either the distribution or the revenue function (e.g., Assumption C.2).

Assumption 2.2. (Lipschitz continuity) The noise distribution F is Lipschitz continuous with a positive constant L, i.e., $|F(x) - F(y)| \leq L|x - y|, \forall x, y \in \mathbb{R}$.

These assumptions are satisfied by a broad range of distributions, such as the uniform distribution and truncated Gaussian.

Under our framework, the seller sequentially observes covariates x_t and dynamically sets prices p_t based on the historical record $\{x_1, p_1, y_1, \ldots, x_{t-1}, p_{t-1}, y_{t-1}, x_t\}$. The seller's informational constraints necessitate balancing two competing objectives: exploration (gathering data to improve estimates of v^* and F) versus exploitation (leveraging current knowledge to set revenue-maximizing prices). Our policy addresses this trade-off by decoupling the learning components: valuation estimation via controlled exploration and revenue optimization through confidence-bound mechanisms. This structure enables simultaneous market learning and near-optimal pricing, without requiring restrictive assumptions about the learning objectives.

3. Algorithm Design

In this section, we propose Algorithm 1, a distribution-free pricing policy for contextual dynamic pricing with unknown v^* and F. Our method uses an episode-based framework with three phases per episode: pure exploration, estimation, and UCB. To achieve minimax optimality, the algorithm sets phase durations based on the horizon length T. When T is unknown, we apply the doubling trick to partition the horizon into exponentially growing episodes. This adaptive strategy drives the policy toward near-optimal performance as more data becomes available.

Our algorithm leverages a novel combination of techniques to address the challenges of dynamic pricing under uncertainty. First, our algorithm ensures that the number of candidate prices within each episode remains bounded. This allow us to employ a tighter Azuma's inequality-based concentration bound instead of the commonly used elliptical potential lemma, which is crucial for controlling regret and achieving minimax optimality under uncertainty. To satisfy the assumption of Azuma's inequality, we adopt a layered data partitioning technique that isolates data across layers, ensuring that the confidence intervals are constructed from statistically independent data.

3.1. Estimation of the valuation function v^*

The seller collects binary feedback y_t, which depends jointly on the posted price p_t, the valuation $v^*(x_t)$, and the unknown noise distribution F. This coupling complicates direct estimation of v^*, as traditional methods would require simultaneous identification of F.

To decouple these components, we adopt a uniform random pricing strategy $p_t \sim \mathrm{Unif}(0, B)$ during exploration phases. This approach, inspired by (Javanmard and Nazerzadeh 2019), induces a regression structure that isolates v^* estimation from the unknown distribution F. Specifically, using uniform pricing over $[0, B]$ leads to the following critical relationship:

$$\mathbb{E}[By_t|x_t] = B\mathbb{E}[\mathbb{E}[\mathbf{I}_{\{p_t \le v^*(x_t)+\epsilon_t\}}|x_t, \epsilon_t]|x_t] = B\mathbb{E}\Big[\frac{v^*(x_t)+\epsilon_t}{B}\Big|x_t\Big] = v^*(x_t)$$

Algorithm 1 Distribution-Free Dynamic Pricing Algorithm

Input: price upper bound B, function space $\mathcal{V}$, confidence parameter δ, and an **Offline Least Square Oracle**

for $k = 1, 2, \cdots$ **do**

  Set the length of the k-th episode as $\ell_k = 2^{k-1}$

  Set the length of the exploration phase at k-th episode as $T_k^e = \left\lceil (k\ell_k)^{\frac{2}{3}} \ln^{\frac{1}{3}}(k|\mathcal{V}|/\delta) \right\rceil$

  for $t = 2^{k-1}+1, \cdots, 2^{k-1} + \min\{T_k^e, \ell_k\}$ **do** ▷ Exploration phase

    Observe context x_t

    Set a price p_t uniformly from $(0, B)$

    Observe the feedback y_t

  if $T_k^e > \ell_k$ **then**

    Continue

  Find the estimator by the **Offline Least Square Oracle** ▷ Estimation phase

$$\hat{v}_k = \arg\min_{v \in \mathcal{V}} \sum_{t=2^{k-1}+1}^{2^{k-1}+T_k^e} (v(x_t) - By_t)^2$$

  Set the length $T_k = \ell_k - T_k^e$

  Set the discretization number $N_k = \left\lceil T_k^{\frac{1}{3}} / \ln^{\frac{2}{3}}(T_k/\delta) \right\rceil$

  for $t = 2^{k-1} + T_k^e + 1, \cdots, 2^k$ **do** ▷ UCB phase

    Apply the **UCB-LDP** (Algorithm 2) on the coming sequential contexts x_t with the estimator $\hat{v}_k$, the discretization number N_k, the length T_k and the bound B

where the noise term ϵ_t vanishes due to its zero-mean property. This enables consistent estimation of v^* through offline regression on exploration-phase data, independent of F. For linear valuation models, standard least squares suffices. General function spaces require corresponding regression oracles, as detailed in Section 5.

To ensure minimax optimality, we set the exploration length $T_k^e = \left\lceil \ell_k^{\frac{2}{3}} \ln^{\frac{1}{3}}(|\mathcal{V}|/\delta) \right\rceil$, balancing estimation precision with the episode duration ℓ_k. This choice ensures minimax optimality in terms of T and $|\mathcal{V}|$. Early episodes with $k \le k^* = \mathcal{O}(\ln\ln(\delta^{-1}|\mathcal{V}|\ln T))$ use pure uniform pricing to guarantee $T_k^e \le \ell_k$.

We first analyze the case where the function space is finite, i.e., $|\mathcal{V}| < \infty$. The expected estimation error in this scenario is presented in Lemma 3.1.

Lemma 3.1. With probability $1-\delta$, the estimator $\hat{v}_k$ in Algorithm 1 satisfies

$$\mathbb{E}[(\hat{v}_k(x)-v^*(x))^2] \leq \frac{18B\ln\left(|\mathcal{V}|\log_2(T_k^e)/\delta\right)}{T_k^e}.$$

While our initial analysis targets the finite-space case, it can be naturally extended to infinite function spaces using metric entropy arguments, as discussed in Section 5. Our framework requires only an upper bound on the mean-squared error of the prediction oracles, which can then be converted to a regret bound using the analysis in Section 4. This flexibility allows our approach to accommodate both general offline regression oracles and classification oracles, highlighting its broad applicability in diverse settings.

Remark 3.1. In Algorithm 1, the valuation function v^* is estimated using only the samples from the current episode k. While it is possible to leverage all prior exploration data, doing so does not improve the final regret bound, as each exploration phase provides a sample size of $\mathcal{O}(\ell_k^{\frac{2}{3}}\ln^{\frac{1}{3}}(|\mathcal{V}|/\delta))$. For clarity, we use only the current episode's data to obtain $\hat{v}_k$, and focus the estimation on a single batch to simplifying the algorithm's presentation.

3.2. Discretization for noise distribution F

During the UCB phase of episode k, we leverage the regression estimator $\hat{v}_k$ to guide a UCB procedure that balances F-learning with revenue maximization. To efficiently explore optimal prices $p\in(0,B)$, we focus on F-values within the range of $p-\hat{v}_k(x)$. Specifically, we restrict F-learning to the interval $[-\|\hat{v}_k\|_\infty, B+\|\hat{v}_k\|_\infty]$ by discretizing it into N_k sub-intervals and concentrating on their midpoints $\{m_j\}_{j\in[N_k]}$. Unlike Chen et al. (2024), Fan et al. (2024), who estimate the entire cumulative distribution function F using the Nadaraya-Watson kernel estimator, our approach simplifies the estimation task by targeting a limited set of F-values. By focusing on a limited number of points, we avoid the complexity of estimating F or possibly F' over the entire domain. This allows us to ensure that our estimation is more focused, with tighter concentration bounds at each point, leading to improved regret bounds.

By focusing on the N_k prices $\{m_j+\hat{v}_k(x_t)\}_{j=1}^{N_k}$ at round t, our formulation exhibits a linear bandit structure. Define the parameter vector

$$\xi_t = \left(1-F(m_1+v^*(x_t)-\hat{v}_k(x_t)), 1-F(m_2+v^*(x_t)-\hat{v}_k(x_t)), \cdots, 1-F(m_{N_k}+v^*(x_t)-\hat{v}_k(x_t))\right).$$

Consider arms a_j defined as a vector of the form $(0,0,\cdots,p_j,\cdots,0)$, containing a single non-zero element $p_j=m_j+\hat{v}_k(x_t)$ at the j-th position. If arm j (corresponding to a price p_j) is played at time t, the revenue can be expressed as $\xi_t^\top a_j$.

This formulation is inspired by Luo et al. (2024), who introduced a perturbed linear bandit problem where $\xi^* = (1 - F(m_1), 1 - F(m_2), \ldots, 1 - F(m_{N_k}))$ is the central parameter vector. While each ξ_t deviates from ξ^*, its magnitude remains controlled due to the Lipschitz continuity in Assumption 2.2 and the error bounds for $\hat{v}_k$ as provided in Lemma 3.1. However, Luo et al. (2024) employ the elliptical potential lemma to construct the UCB, which results in a suboptimal regret upper bound, leaving room for further improvement in terms of regret optimization.

For further optimization, a key observation is that the number of actions in the k-th episode is bounded by N_k. In such scenario, the linear contextual bandits literature such as Auer (2002), Li et al. (2019) has demonstrated that the elliptical potential lemma is not strictly necessary. These papers propose a layered data partitioning method, which can further improve the regret value under linear rewards.

However, this method is not directly applicable to our dynamic pricing setup. The first challenge arises from parameter normalization: when we normalize ξ^* to apply the layered data partitioning method, its ℓ_2 norm is $\mathcal{O}(\sqrt{N_k})$, which introduces an additional $\sqrt{N_k}$ term into the regret bound. This additional $\sqrt{N_k}$ factor leads to a larger regret bound, since the number of discretization points, N_k, typically grows polynomially with T.

To avoid this, we exploit the fact that each action vector has only one non-zero component and satisfies $|(m_j + \hat{v}_k(x_t))(1 - F(m_j))| \leq B$. This property allows us to apply a component-wise UCB using ℓ_∞ estimation error instead of the ℓ_2 error, thus bypassing the extra $\sqrt{N_k}$ term. We can then use Azuma's inequality, rather than the elliptical potential lemma, to obtain concentration bounds. Our algorithm incorporates a layered data partitioning strategy (Auer 2002, Chu et al. 2011), which ensures the statistical independence needed for Azuma's inequality.

A second challenge arises from the perturbation of ξ^*, which introduces misspecification errors in the reformulated linear bandit. Specifically, for a price p_t and covariate x_t, we have

$$\mathbb{E}[y_t | x_t, p_t = m_j + \hat{v}_k(x_t)] = 1 - F(m_j + \hat{v}_k(x_t) - v^*(x_t)).$$

This is an approximation of $\xi_j^* = 1 - F(m_j)$ that holds exactly only if $\hat{v}_k$ is error-free. As per Assumption 2.2, the magnitude of perturbations scales with the accuracy of the estimator. When $\hat{v}_k$ is exact, the problem reverts to a classic linear bandit where methods like Auer (2002), Li et al. (2019) apply directly. Conversely, larger estimation errors amplify this perturbation, increasing both the difficulty and the potential regret, as shown in Theorem 4.1. This situation calls for a more nuanced use of layered data partitioning under model misspecifications.

3.3. Layered data partitioning

For the balance between revenue maximization and F-learning in the UCB phase, we propose the UCB-LDP algorithm (Algorithm 2). In particular, we employ a layered data partitioning technique to address the challenge of statistical dependencies in sequential decision-making while maintaining tight confidence bounds. The idea is to organize data from past rounds into layers with different confidence levels, which introduces independent between each layer's estimators and allows us to apply simpler concentration bounds.

Algorithm 2 Upper Confidence Bound with Layered Data Partitioning (**UCB-LDP**)

Input: the estimator $\hat{v}$, the discretization number N, the length T and the bound B

Divide the F-learning interval $[-\|\hat{v}\|_\infty, B+\|\hat{v}\|_\infty]$ into N equal-length intervals with their midpoints denoted as $m_1, \cdots, m_N$

Let $S = \left\lceil \log_2(\sqrt{T}) \right\rceil$, $\Psi_t^s = \emptyset$ for $s \in [S]$

for $t = 1, 2, \cdots, T$ **do**

  Observe the context x_t

  Let $s = 1$ and $\mathcal{A}_{t,1} = \{j \in [N] | m_j + \hat{v}(x_t) \in (0, B)\}$

  repeat

    Compute $\Psi_t^s(j) = \{\tau \in \Psi_t^s | m_j + \hat{v}(x_\tau) = p_\tau\}$ for all $j \in [N]$

    For all $j \in [N]$, compute confidence radius $r_{t,s}^j$ defined in (1) and $\mathrm{UCB}_{t,s}^j$ defined in (2)

    if $(m_j + \hat{v}(x_t)) r_{t,s}^j \le B/\sqrt{T}$ for **all** $j \in \mathcal{A}_{t,s}$ **then** ▷ Exploitation

      Choose the index $j_t = \arg\max_{j \in \mathcal{A}_{t,s}} \mathrm{UCB}_{t,s}^j$

      Keep $\Psi_{t+1}^\sigma = \Psi_t^\sigma$ for $\sigma \in [S]$

    else if $(m_j + \hat{v}(x_t)) r_{t,s}^j \le B 2^{-s}$ for **all** $j \in \mathcal{A}_{t,s}$ **then** ▷ Price elimination

      Let $\mathcal{A}_{t,s+1} = \{j \in \mathcal{A}_{t,s} | \mathrm{UCB}_{t,s}^j \ge \max_{j' \in \mathcal{A}_{t,s}} \mathrm{UCB}_{t,s}^{j'} - B 2^{1-s}\}$

      Advance to the next layer $s \leftarrow s + 1$

    else ▷ Exploration

      Choose j_t such that $(m_j + \hat{v}(x_t)) r_{t,s}^j > B 2^{-s}$

      Set the stopping layer $s_t = s$

      Update $\Psi_{t+1}^{s_t} = \Psi_t^{s_t} \cup \{t\}$ and keep $\Psi_{t+1}^\sigma = \Psi_t^\sigma$ for $\sigma \neq s_t$

  until An index j_t is found

  Set the price $p_t = m_{j_t} + \hat{v}(x_t)$

The core idea of layered data partitioning is to maintain adaptive statistical precision through layers, with lower layers corresponding to the widest confidence bound. In particular, the algorithm organizes the data into layers $s \in [S]$, where each layer corresponds to a different level of precision for price optimization. For epoch k, we set the number of layers as $S = \lceil \log_2(\sqrt{T_k}) \rceil$. At time t, each layer s maintains its own dataset Ψ_t^s, containing rounds prior to time t that made their pricing decisions at layer s. As a result, the historical data $[t-1]$ is partitioned into disjoint sets $\{\Psi_t^s : s \in [S]\}$. This isolation ensures that confidence intervals computed for layer s depend only on data from s avoiding cross-layer dependencies and enabling the use of Azuma's inequality for tight concentration bounds.

At time t, the data of each round belongs to exactly one *stopping layer* s_t, determined alongside the pricing decision p_t. The algorithm starts at layer $s = 1$ and examines successive layers until it selects a price. While visiting the layers, we maintain a set of candidate prices $\mathcal{A}_{t,s}$ with $\mathcal{A}_{t,1} = \{j \in [N_k] | m_j + \hat{v}_k(x_t) \in (0, B)\}$ including all discretized prices in $(0, B)$. As we move through layers, the set of candidate prices $\mathcal{A}_{t,s}$ may shrink: $\mathcal{A}_{t,1} \supseteq \mathcal{A}_{t,2} \supseteq \cdots \supseteq \mathcal{A}_{t,S}$. For each price $p_j = m_j + \hat{v}_k(x_t)$, let $\Psi_t^s(j) \subset \Psi_t^s$ be the set of previous rounds in layer s that used p_j. At time t and layer s, we estimate ξ_j^* via the sample mean

$$w_{t,s}^j = \frac{1}{|\Psi_t^s(j)|} \sum_{\tau \in \Psi_t^s(j)} y_\tau$$

and condifence radius

$$r_{t,s}^j = \min\left\{ \sqrt{\frac{\ln(N_k T_k/\delta)}{2|\Psi_t^s(j)|}}, 1 \right\}. \tag{1}$$

We consequently compute the UCB for the revenue of price p_j at round t and layer s as

$$\mathrm{UCB}_{t,s}^j \triangleq (m_j + \hat{v}(x_t))(w_{t,s}^j + r_{t,s}^j). \tag{2}$$

If $\Psi_t^s(j)$ is empty, indicating no previous round has selected price p_j, we set its UCB to infinity.

The algorithm's progression through layers is guided by the level of revenue uncertainty, determined by $(m_j + \hat{v}_k(x_t))r_{t,s}^j$:

(a) **Exploitation**: If the revenue uncertainty is small for *all* prices indexed in $\mathcal{A}_{t,s}$, as specified by $(m_j + \hat{v}_k(x_t))r_{t,s}^j \le B/\sqrt{T_k}$, the algorithm selects the price p_j with the highest UCB, stops layer traversal, and keeps the datasets.

(b) **Price elimination**: If *all* prices indexed in $\mathcal{A}_{t,s}$ satisfy $(m_j + \hat{v}_k(x_t))r_{t,s}^j \le B \cdot 2^{-s}$ and some prices satisfy $(m_j + \hat{v}_k(x_t))r_{t,s}^j > B/\sqrt{T_k}$, we eliminate those whose UCB is worse than the maximum by at least $B2^{1-s}$, and continue to the next layer with the remaining prices.

(c) **Exploration**: If there exists a price p_t such that $(m_j + \hat{v}_k(x_t))r^j_{t,s} > B \cdot 2^{-s}$, the uncertainty is substantial; the algorithm selects any such price for further exploration, stops layer traversal, and assigns round t to the stopping layer $s_t = s$.

Because the stopping layer s_t at round t depends only on $\{\Psi^s_t\}_{s \le s_t}$, the statistical correlations across layers are decoupled. This property is formally analyzed by Li et al. (2019), Auer (2002) for linear bandits. The following lemma provides a key result on the UCB property of the algorithm.

Lemma 3.2. Fix an episode k. For any round t in the episode k, and any layer s at round t, with probability at least $1 - \frac{\delta}{N_k T_k}$, each $j \in [N_k]$ satisfies

$$|\xi^*_j - w^j_{t,s}| \le r^j_{t,s} + L\eta^j_{t,s}, \quad \text{where} \quad \eta^j_{t,s} = \frac{1}{|\Psi^s_t(j)|} \sum_{\tau \in \Psi^s_t(j)} |\hat{v}_k(x_\tau) - v^*(x_\tau)|$$

Based on Lemma 3.2, we construct a high-probability event

$$\Gamma_k = \left\{ |\xi^*_j - w^j_{t,s}| \le r^j_{t,s} + L\eta^j_{t,s}, \forall t, \forall s, \forall j \right\}. \tag{3}$$

We see that $\mathbb{E}_x[\eta^j_{t,s}] = \mathbb{E}_x[|\hat{v}_k(x) - v^*(x)|]$ for any t, s, j, under the i.i.d. assumption for x_t. Combining with a union bound implies that

$$\mathbb{P}(\overline{\Gamma_k}) \le \frac{\delta}{N_k T_k} \times N_k \times T_k \times S = \delta S = \delta \left\lceil \log_2 \left(\sqrt{T_k} \right) \right\rceil.$$

The UCB property offers an intuitive decomposition: the first term captures the variance from random feedback y_t given x_t and p_t, and the second term represents the bias from estimating $\hat{v}_k$. As episodes progress, more data accumulate, reducing both variance and bias, so that $w^j_{t,s}$ converges to ξ^*_j. Crucially, our construction implies that the resulting bounds do not explicitly depend on the (unknown) Lipschitz constant for F. This follows from the fact that comparing UCB values relative to each other cancels out the bias term, which is common across all indices. Hence, our approach sidesteps the need for explicit knowledge of Lipschitz parameters, further simplifying implementation.

4. Regret Analysis

In this section, we analyze the regret of our proposed Algorithm 1. We first analyze a single episode of Algorithm 1 and then extend the analysis to the entire horizon.

4.1. Upper bounds

During the initial pure exploration phase, we obtain an estimator $\hat{v}_k$. Since this estimator guides the UCB procedure, it is important to understand how its error affects the regret in the UCB phase. A lower estimation error is expected to lead to lower regret. However, reducing the error requires a longer exploration phase, which in turn increases regret. Thus, besides the "inner" balance between revenue maximization and F-learning in the UCB phase, there is also an "outer" balance between the regret incurred during the exploration phase and that incurred in the UCB phase, both of which relate to learning v^*.

For each round t in a fixed episode k, we denote the best price from the discretized set as

$$\tilde{p}_t^* \triangleq m_{j_t^*} + \hat{v}_k(x_t), \quad \text{where} \quad j_t^* = \arg\max_{j \in [N_k]} \mathsf{Rev}_t\left(m_j + \hat{v}_k(x_t)\right).$$

In our regret analysis, we decompose the per-round regret into two parts: the learning and the discretization regret. In particular, the regret at round t can be written as

$$\mathsf{Rev}_t(p_t^*) - \mathsf{Rev}_t(p_t) = \underbrace{\mathsf{Rev}_t(\tilde{p}_t^*) - \mathsf{Rev}_t(p_t)}_{\mathcal{R}_t^1} + \underbrace{\mathsf{Rev}_t(p_t^*) - \mathsf{Rev}_t(\tilde{p}_t^*)}_{\mathcal{R}_t^2}.$$

The term $\mathcal{R}_t^1$ is called the *learning regret*; it measures the loss incurred by the estimation error in the unknown valuation functions v^* and learning F. The term $\mathcal{R}_t^2$ is called the *discretization regret*; it quantifies the loss due to discretizing the continuous price decision.

Learning regret. We first analyze the learning regret, denoted as $\sum_{t=1}^T \mathcal{R}_t^1$, from the inner UCB algorithm while keeping the exploration and estimation phases fixed.

Define the discrete best price at layer s as

$$\tilde{p}_{t,s}^* \triangleq m_{j_{t,s}^*} + \hat{v}_k(x_t), \quad \text{where} \quad j_{t,s}^* = \arg\max_{j \in \mathcal{A}_{t,s}} \mathsf{Rev}_t(m_j + \hat{v}_k(x_t)).$$

If the estimation of $\hat{v}_k$ is exact, our algorithm ensures $j_{t,s}^* = j_t^*$ under the high probability event Γ_k in (3), and hence classical results in Auer (2002) can be applied, because the optimal action will not be eliminated when Γ_k holds.

However, the estimation error in $\hat{v}_k$ may result in the undesired elimination of the best discretized price for each layer s, leading to a revenue gap that propagates through the layers. Specifically, we want to bound the revenue difference between $\tilde{p}_{t,s}^*$ and $\tilde{p}_t^*$. Fortunately, this gap can be controlled with careful analysis. When advancing to a new layer, the revenue difference between the prices indexed by $j_{t,s}^*$ and $j_{t,s+1}^*$ is naturally proportional to the misspecification error, which can be upper bounded by $BL\mathbb{E}_x\left[|\hat{v}_k(x) - v^*(x)|\right]$. For any layer $s \in [s_t - 1]$, such error propagation occurs $s-1$ times, and we can prove that the revenue gap is upper bounded by $4BL\mathbb{E}_x[|\hat{v}_k(x) - v^*(x)|](s-1)$, as stated in Lemma 4.1.

Lemma 4.1. For each round t in episode k and each layer $s \in [s_t - 1]$, under the high probability event Γ_k, we have

$$\mathbb{E}_{x_t}\left[\mathsf{Rev}_t\left(\tilde{p}_t^*\right) - \mathsf{Rev}_t\left(\tilde{p}_{t,s}^*\right)\right] \leq 4BL\mathbb{E}_x\left[|\hat{v}_k(x) - v^*(x)|\right](s-1).$$

Now, we can effectively control the regret within layer s using $j_{t,s}^*$ as the benchmark. The condition stated in Step 9 of Algorithm 2 guarantees that the confidence radius of prices corresponding to their indexes in $\mathcal{A}_{t,s}$ exponentially decreases as s increases. The additional bias term in Lemma 4.2 arises from the parameter perturbation.

Lemma 4.2. For each round t in episode k and each layer $2 \leq s \leq s_t - 1$, under the high probability event Γ_k, we have

$$\mathbb{E}_{x_t}\left[\mathsf{Rev}_t\left(\tilde{p}_{t,s}^*\right) - \mathsf{Rev}_t\left(p_t\right)\right] \leq 8B \cdot 2^{-s} + 4BL\mathbb{E}_x\left[|\hat{v}_k(x) - v^*(x)|\right].$$

Combining Lemma 4.1 and Lemma 4.2, we immediately have an upper bound on the learning regrets for prices in layer s. The upper bound in Lemma 4.3 can be divided into two components: the variance and the bias. As we increase the value of s, the variance decreases exponentially, while the bias only increases linearly. Therefore, having a larger value of s is advantageous for online learning.

Lemma 4.3. For each round t in episode k and each layer $2 \leq s \leq s_t - 1$, under the high probability event Γ_k, we have

$$\mathbb{E}_{x_t}\left[\mathsf{Rev}_t\left(\tilde{p}_t^*\right) - \mathsf{Rev}_t\left(p_t\right)\right] \leq 8B \cdot 2^{-s} + 4BL\mathbb{E}_x[|\hat{v}_k(x) - v^*(x)|]s.$$

To obtain an upper bound on the cumulative regret, we decompose the time horizon into epochs, and further, into layers within each epoch. The cumulative regret over all rounds in Ψ_t^s can be bounded by summing the regret for each round in Ψ_t^s, which is effectively bounded by a function of the layer index s, as shown in Lemma 4.3. In order to bound the cumulative regret over all rounds in Ψ_t^s, we need to determine the size of $\Psi_{T_k+1}^s$ (see Lemma C.3). Thus, the overall cumulative learning regret of the inner UCB algorithm is bounded by the sum of the regret over all layers and all epochs. Formally, we establish the following lemma to bound the learning regret of the inner UCB algorithm.

Lemma 4.4. Under Assumption 2.1 and Assumption 2.2, the expected learning regret $\mathbb{E}\left[\sum_{t=1}^T \mathcal{R}_t^1\right]$ is upper bounded by

$$\sum_{k=1}^{\lceil \log_2 T \rceil} \left(8B\sqrt{2N_k T_k \ln(T_k N_k/\delta)\ln T_k} + 4BL\ln^{\frac{1}{3}}(|\mathcal{V}|\ln T/\delta)(k\ell_k)^{\frac{2}{3}} + 2B\sqrt{T_k} + 8N_k\ln(T_k N_k/\delta)\right)$$

with probability at least $1-\delta$.

The regret bound in Lemma 4.4 indicates that a higher level of estimation accuracy for both F and v^* leads to a lower regret in the UCB phase. The first term in the regret bound captures the impact of uncertainty in estimating the discretized F-values. The second term in arises from the estimation error of $\hat{v}_k$, i.e., the misspecification error, in the linear bandit problems. The last two terms are of order of $\mathcal{O}(\sqrt{T})$ and dominated by the first two terms.

Remark 4.1. Luo et al. (2022) introduced a perturbed linear bandit problem to model dynamic pricing and obtained an expected learning regret bound of $\mathcal{O}(N\sqrt{T\ln T})$. Compared with their work, we improve the bound to $\mathcal{O}(\sqrt{NT\ln T\ln N})$ by using the ℓ_∞ norm to bound the revenue. In particular, the revenue at round t takes the form $\xi_t^\top a_j$ and we may consider either the ℓ_2 norm or the ℓ_∞ norm to bound the revenue:

$$\xi_t^\top a_j \le \|\xi_t\|_2 \|a_j\|_2 \le B\|\xi_t\|_2 \quad \text{or} \quad \xi_t^\top a_j \le \|\xi_t\|_\infty \|a_j\|_1 \le B\|\xi_t\|_\infty .$$

The ℓ_∞ bound exploits the fact that each action vector has exactly one non-zero component, and avoids an additional $\sqrt{N}$ term compared to the bound obtained using the ℓ_2 norm in Luo et al. (2022). However, switching from the ℓ_2 norm to the ℓ_∞ norm is not trivial. To address this, we utilize Azuma's inequality instead of the elliptical potential lemma to derive concentration bounds. Additionally, we apply the layered data partitioning technique to meet the requirements for Azuma's inequality. A critical insight that underpins our approach is that the number of candidate prices after discretization is finite. This allows us to establish an ℓ_∞-based UCB for each candidate price. If the number of prices were infinite and the revenue function lacked structure, it would be difficult to apply an ℓ_∞-based UCB.

Discretization regret. We now analyze the discretization regret, denoted as $\sum_{t=1}^T \mathcal{R}_t^2$, which arises from the discretization of the continuous price space.

Intuitively, a denser discretization, achieved by increasing the number of price points, leads to a reduction in discretization regret because a finer grid of prices brings the discrete best price closer to the overall best price. This intuition is formalized by Lemma 4.5, which establishes that the discretization regret is inversely proportional to the number of candidate prices N_k and proportional to the number of rounds T_k.

Lemma 4.5. Under Assumption 2.1, the expected discretization regret $\mathbb{E}\left[\sum_{t=1}^T \mathcal{R}_t^2\right]$ is upper bounded by

$$\mathbb{E}\left[\sum_{t=1}^T \mathcal{R}_t^2\right] \le \sum_{k=1}^{\lceil \log_2 T\rceil} \frac{4BT_k}{N_k}.$$

Regret upper bound. To achieve an overall minimized regret, we need to balance the learning and discretization regrets optimally. In particular, increasing the discretization parameter N_k results in a lower discretization regret but a higher learning regret, as indicated by Lemma 4.4 due to the increased number of candidate prices and the resulting complexity in the search process. We achieve this balance by setting $N_k = \tilde{\mathcal{O}}(T_k^{\frac{1}{3}})$, which leads to a $\tilde{\mathcal{O}}(T^{\frac{2}{3}})$ regret upper bound.

Theorem 4.1. Under Assumption 2.1 and Assumption 2.2, the expected regret satisfies

$$\mathbb{E}[\mathrm{Reg}(T)] = \mathcal{O}\left(BT^{\frac{2}{3}} \ln^{\frac{1}{3}}\left(\delta^{-1}|\mathcal{V}| \ln T\right) \ln^{\frac{2}{3}}\left(\delta^{-1} T \ln T\right)\right)$$

with probability at least $1-\delta$.

We remark that our regret bound depends on $\mathcal{V}$ only through the expected estimation error from the offline regression oracle tailored to the finite function space; see Lemma 3.1. In Section 5, we provide extensions to incorporate general function spaces and discuss their implications on the regret upper bound, where we also compare our results with existing works under the linear valuation model.

4.2. Lower bounds

Xu and Wang (2022) establish a regret lower bound of $\Omega(T^{\frac{2}{3}})$ for the non-contextual pricing problem with a Lipschitz continuous noise distribution F. Combined with the $\tilde{\mathcal{O}}(T^{\frac{2}{3}})$ regret upper bound in Theorem 4.1, the minimax optimality of Algorithm 1 up to logarithmic terms is established.

Furthermore, we present a general lower bound for the non-contextual pricing problem with an m-th differentiable noise distribution F in Theorem 4.2. Several important special cases are in order. When $m=1$, Lipschitz continuity holds for F, and we recover the $\Omega(T^{\frac{2}{3}})$ lower bound in Xu and Wang (2022). The case $m=2$ implies Lipschitz continuity and second-order smoothness in F, which aligns with the $\Omega(T^{\frac{3}{5}})$ regret lower bound in Luo et al. (2022). Moreover, since our constructed hard instances satisfy the requirements in Wang et al. (2021), our results also imply an $\Omega(T^{\frac{m+1}{2m+1}})$ lower bound for a general m-th differentiable demand function.

Theorem 4.2. Consider a non-contextual pricing problem where the valuations y_t are independently and identically generated from an unknown distribution. Let the distribution satisfy the following conditions:

1. $F(\cdot)$ is m times differentiable on a bounded interval $[a,b]$.

2. The revenue function $\mathsf{Rev}(x) = x(1-F(x))$ has a unique maximizer within the interval $[a,b]$.

Then, no policy can achieve an $\mathcal{O}\left(T^{\frac{m+1}{2m+1}-\delta}\right)$ regret bound for any $\delta > 0$, where T represents the number of pricing rounds.

Proof sketch. To establish our lower bounds, we employ techniques similar to those in Luo et al. (2022), Wang et al. (2021), Xu and Wang (2022). Note that Wang et al. (2021) construct an m-th differentiable demand function; however, their construction does not satisfy the requirements of a valid cumulative distribution function. To address this gap, we construct a valid distribution that meets the conditions outlined in Theorem 4.2.

We begin by considering a non-negative, infinitely differentiable function $u_0(x)$ that vanishes outside a bounded interval. Integrating $u_0(x)$ yields a non-decreasing function $u(x)$. Based on $u(x)$, we construct a sequence of "bump towers" $B_{[a_k,b_k]}(x)$ that possess the desired properties. In particular, for each interval $[a_k, b_k]$ with length $w_k = 3^{-k!}$, we partition it into three equally sized sub-intervals. The middle sub-interval is then subdivided further into candidate intervals, each of which is used to form the next interval $[a_{k+1}, b_{k+1}]$. By summing the bump functions defined on these nested intervals, we obtain the bump tower. In contrast to the constructions in Luo et al. (2022), Xu and Wang (2022), our approach uses coefficients that scale with w_k^m in the function series representation, ensuring that the resulting distribution satisfies the m-th order smoothness assumption.

Finally, we show that any policy will frequently miss the peaks of certain revenue functions, which leads to an inevitable accumulation of regret. This accumulation arises from the policy's inability to fully capture the potential revenue due to the carefully constructed structure of the distribution functions.

Remark 4.2. The Lipschitz continuity assumption is crucial as it enables us to control the misspecification error in the regret analysis. Without the Lipschitz condition, Xu and Wang (2022) proposed an algorithm with a regret bound of $\tilde{\mathcal{O}}(T^{\frac{3}{4}})$. In their work, they discretize the linear parameter space and distribution space to form the policy space, reducing the original problem into an EXP4 agent. Although they discretize the distribution in a similar way to us, they still consider finding a good distribution within the given set, resulting in a policy space of size $\mathcal{O}(2^{N_k})$. In contrast, for setting a suitable price, we only need to evaluate at N_k points. This observation effectively reduces the search space from $\mathcal{O}(2^{N_k})$ to $\mathcal{O}(N_k)$, leading to polynomial and exponential decrease in regret and time complexity, respectively. However, it is currently unknown whether their bound is tight in the absence of the Lipschitz continuity assumption. Further investigation is required to explore the tightness of the regret bound without the Lipschitz assumption. Additionally, there remains a gap of order $|\mathcal{V}|$ in terms of lower bounds, which is an area for future research.

Remark 4.3. Fan et al. (2024) propose a dynamic pricing policy for m-th differentiable F. While their work is insightful, their regret upper bound scales as $\tilde{\mathcal{O}}(T^{\frac{2m+1}{4m-1}})$, which, as shown in Theorem 4.2,

leaves room for improvement. Similarly, Chen et al. (2024) study m-th differentiable F under a general valuation model and achieve regret upper bounds of $\tilde{\mathcal{O}}\left(T^{\frac{2m+1}{4m-1}\vee\frac{d_0+4}{d_0+8}}\right)$. Notably, a gap persists between these upper bounds and our derived lower bound. This gap arises for two possible reasons. First, their policy might not achieve optimality, as evidenced by the $\tilde{\mathcal{O}}(T^{\frac{3}{4}})$ regret upper bounds they incur when F is only Lipschitz, whereas a $\tilde{\mathcal{O}}(T^{\frac{2}{3}})$ regret upper bound is attainable in such cases. Second, the inherent uncertainty in valuation models could increase the problem's complexity, potentially leading to a larger lower bound than what we derived. As a result, designing a minimax optimal policy for smoother distributions remains an open problem.

Remark 4.4. To intuitively understand the coefficients of parameters in the regret lower bound, consider a simplified scenario with a d_0-dimensional linear valuation model. Suppose we design d_0 distinct challenging sub-problems, each corresponding a unique context. Under a uniform context distribution, each sub-problem appears approximately T/d_0 times over the horizon T. For each sub-problem, the regret lower bound scales as $\Omega((T/d_0)^{\frac{2}{3}})$, which is the typical regret rate of non-contextual dynamic pricing problems. Aggregating the regret across all d_0 sub-problems yields a total lower bound of $\Omega(d_0^{\frac{1}{3}}T^{\frac{2}{3}})$. This illustrates how the interplay between dimension and time horizon arises: the learner must resolve uncertainty across contexts while adapting to demand shifts, further complicating the task of exploration.

5. Extensions

In this section, we explore several extensions to our framework, aiming to enhance our understanding of the theoretical results and their applications.

5.1. General offline regression oracle for general function space

Our proof techniques rely solely on the estimation error assumption of a regression oracle (see Lemma 3.1). This enables us to extend our results to a general offline regression oracle that can be applied to various function spaces.

To this end, we assume that a general offline regression oracle is available to generate an estimator of the unknown value function $v^* \in \mathcal{V}$ based on the observed data.

Assumption 5.1. The general offline regression oracle takes n samples of the form (x_i, y_i), which are independently and identically drawn according to a fixed distribution such that $\mathbb{E}[By_i \mid x_i] = v^*(x_i)$ for $v^* \in \mathcal{V}$. It returns a predictor $\hat{v} \in \mathcal{V}$ satisfying that for any $\delta > 0$,

$$\mathbb{E}_x\left[\left(\hat{v}(x) - v^*(x)\right)^2\right] \leq \frac{\rho_{\mathcal{V}}(\delta)}{n},$$

with probability at least $1-\delta$.

This assumption quantifies the estimation accuracy in statistical learning. The term $\rho_{\mathcal{V}}(\delta)$ reflects the intrinsic complexity of learning the function space $\mathcal{V}$ as we decrease δ. In particular, $\frac{\rho_{\mathcal{V}}(\delta)}{n}$ bounds the expected squared L^2 error between v^* and $\hat{v}$. Under the realizability assumption ($v^* \in \mathcal{V}$), this error precisely measures the estimation error of $\hat{v}$. Obtaining sharp bounds on this error and designing efficient algorithms to achieve them are central in statistical learning tasks.

Example 1 (Finite function space). For the special case where $\mathcal{V}$ is finite, we have $\rho_{\mathcal{V}}(\delta) = \mathcal{O}\left(\ln\left(|\mathcal{V}|/\delta\right)\right)$, as shown in Lemma 3.1. This bound is tight, as it matches the minimax lower bound of $\Omega\left(\ln\left(|\mathcal{V}|/n\right)\right)$ in empirical estimation (Fan 1953).

Assuming the availability of such an offline regression oracle, we extend our results under a general estimation assumption via a simple corollary. By replacing $\ln\left(|\mathcal{V}|/\delta\right)$ with $\rho_{\mathcal{V}}(\delta)$ in the algorithm and proofs, we obtain an extended regret bound. For completeness, we present Algorithm 5 in the Appendix. Specifically, we establish the regret upper bound under the assumptions of boundedness, Lipschitz continuity, and the availability of an offline regression oracle.

Corollary 5.1. Under Assumption 2.1, Assumption 2.2, and Assumption 5.1, the expected regret of Algorithm 5 satisfies

$$\mathbb{E}[\mathrm{Reg}(T)] = \mathcal{O}\left(BT^{\frac{2}{3}}\rho_{\mathcal{V}}^{\frac{1}{3}}(\delta)\ln^{\frac{2}{3}}\left(\delta^{-1}T\ln T\right)\right)$$

with probability at least $1-\delta$.

Application to linear function spaces. In light of the general regret bound in Corollary 5.1, the function class $\mathcal{V}$ can be chosen flexibly to include a variety of parametric or nonparametric models, such as Lipschitz functions, reproducing kernel Hilbert spaces, and neural networks.

To achieve optimal regret in the contextual dynamic pricing problem, it is critical to use an offline regression oracle that is both computationally efficient and statistically optimal for the chosen function class $\mathcal{V}$. To illustrate the versatility of our approach, we focus on linear function spaces commonly employed in the literature (Fan et al. 2024, Luo et al. 2024, Xu and Wang 2022, Golrezaei et al. 2019) and examine the corresponding regret bounds.

In the case of linear valuation models, consider the function space $\mathcal{V} = \{\theta^\top x : \|\theta\|_2 \le 1, \theta \in \mathbb{R}^{d_0}\}$, where d_0 denotes the dimension of the covariates. Under appropriate regularity conditions (e.g., the covariance matrix is positive definite), Assumption 5.1 holds with statistical complexity $\rho_{\mathcal{V}}(\delta) = \mathcal{O}(d_0 \ln(1/\delta))$. This bound is tight, as confirmed by minimax theory (Fan 1953).

Under the additional Lipschitz condition (Assumption 2.2), it follows from Corollary 5.1 that Algorithm 1 achieves a regret of order $\tilde{\mathcal{O}}(d_0^{\frac{1}{3}}T^{\frac{2}{3}})$ in the linear valuation model. This represents a

significant improvement over the previously reported bound of $\tilde{\mathcal{O}}(d_0 T^{\frac{3}{4}})$ in Luo et al. (2024), Xu and Wang (2022). Moreover, when combined with the lower bound in Theorem 4.2, our method is minimax optimal, thus bridging the gap in the existing literature.

Remark 5.1. (High-dimensional covariates with sparsity) Our approach naturally extends to the case of high-dimensional covariates with sparsity s_0. Under mild conditions, Oh et al. (2021) utilizes Lasso regression to obtain an estimator with $\rho_{\mathcal{V}}(\delta) = \mathcal{O}(s_0 \ln(d_0/\delta))$, while Javanmard and Nazerzadeh (2019) employs the regularized maximum likelihood method, resulting in a similar estimator error. Leveraging sparsity, we are able to improve the upper bound on regret to $\tilde{\mathcal{O}}(s_0^{\frac{1}{3}} T^{\frac{2}{3}})$. With similar arguments, we can also extend our results to reproducing kernel Hilbert spaces (Steinwart et al. 2009, Mendelson and Neeman 2010).

Additional smoothness assumption in existing methods. Intuitively, imposing an additional smoothness condition on the expected revenue functions reduces the complexity of dynamic pricing problems. Second-order smoothness of the expected revenue functions around the optimal prices is often imposed in the literature (Luo et al. 2024, Chen and Gallego 2021, Luo et al. 2022). If the revenue function is twice differentiable and the optimal price is an interior point within the price decision range $[0, B]$, then a simple Taylor expansion around the optimal price leads to the following assumption. However, if the revenue function lacks smoothness, this assumption implicitly requires that the optimal price be unique for all covariates.

Assumption 5.2. (Second-order smoothness) Define the general expected revenue function associated with the noise distribution F as $\mathsf{Rev}_q(p) = p\,(1 - F(p - q))$. There exists a positive constant C such that for any $x \in \mathcal{X}$ and $q = v^*(x)$, $\mathsf{Rev}_q\,(p^*(x)) - \mathsf{Rev}_q(p) \le C\,(p^*(x) - p)^2$.

As shown in Theorem 4.2 and by Luo et al. (2022), the dynamic pricing problems satisfying both Assumption 2.2 and Assumption 5.2 have a regret lower bound of at least $\Omega\left(T^{\frac{3}{5}}\right)$, which demonstrates that the second-order smoothness assumption is non-trivial and effectively reduces the difficulty of dynamic pricing problems.

Luo et al. (2022) leverage Assumption 5.2 of the revenue functions to achieve a regret bound of $\tilde{\mathcal{O}}(d_0 T^{\frac{2}{3}})$. In contrast, our method achieves $\tilde{\mathcal{O}}(d_0^{\frac{1}{3}} T^{\frac{2}{3}})$ even without relying on the second-order smoothness assumption.

Fan et al. (2024) study the dynamic pricing problem under the assumption of bounded second derivatives of F (which implies Assumption 5.2) and achieve a regret of $\tilde{\mathcal{O}}\left(T^{\frac{5}{7}}\right)$. Their higher regret upper bound arises because their approach requires estimating F' to determine the price at each round, hence introducing additional error, whereas our method only estimates the values of specific points of F.

5.2. Explicit knowledge of the distribution or valuation

5.2.1. Knowledge of F. When the distribution F is fully known to sellers, Javanmard and Nazerzadeh (2019) propose an algorithm achieving a regret of $\mathcal{O}(d_0 \ln T)$ under smoothness conditions for F. They further show that if the covariance matrix is not positive definite, an alternative method with a regret of $\mathcal{O}(\sqrt{T \ln d_0})$ can be used. Clearly, knowledge of F plays a crucial role in reducing regret. Extending their framework, we integrate the offline oracle assumption (Assumption 5.1) to develop a distribution-dependent algorithm (Algorithm 3). The regret upper bound of Algorithm 3 is summarized in the following corollary.

Corollary 5.2. Assume that the noise distribution F is twice continuously differentiable, and F and $1 - F$ are log-concave. Under Assumption 5.1, the expected regret of Algorithm 3 satisfies

$$\mathbb{E}[\mathrm{Reg}(T)] = \mathcal{O}\left(\rho_{\mathcal{V}}(\delta) \ln T\right)$$

with probability at least $1 - \delta$.

When we relax the differentiability and concavity assumptions, we have the following weaker result under the Lipschitz assumption alone.

Corollary 5.3. Under Assumption 2.2 and Assumption 5.1, the expected regret of Algorithm 3 satisfies

$$\mathbb{E}[\mathrm{Reg}(T)] = \mathcal{O}\left(\sqrt{\rho_{\mathcal{V}}(\delta) T \ln T}\right)$$

with probability at least $1 - \delta$.

Remark 5.2. Cohen et al. (2020) also study the linear valuation model with a known noise distribution. They consider sub-Gaussian noise distributions rather than those satisfying Lipschitz continuity assumption (Assumption 2.2). Their method employs shallow cuts for ellipsoids, retaining over half of the original uncertainty set. Using this technique, they achieve regret guarantees of $\mathcal{O}(d_0^{\frac{19}{6}} T^{\frac{2}{3}} \ln^{\frac{11}{6}} T)$. However, their approach relies on the linearity of the valuation functions, which limits its generalizability. In contrast, our method extends beyond linear models by considering general function spaces and reducing the estimation problem to a general offline regression problem.

5.2.2. Knowledge of v_t. In many practical applications, sellers can directly observe customer valuations v_t (e.g., via bidding). This extra information simplifies the algorithm design by eliminating the need for an exploration phase. In this setting, we modify Algorithm 1 to incorporate the tuple (x_t, v_t) instead of (x_t, y_t) in the offline oracles. Remarkably, when only the Lipschitz continuity assumption (Assumption 2.2) is imposed, the regret of the modified algorithm remains

$\mathcal{O}\left(BT^{\frac{2}{3}}\rho_{\mathcal{V}}^{\frac{1}{3}}(\delta)\ln^{\frac{2}{3}}T\right)$, since the true mean function v^* still needs to be estimated in the presence of noise ϵ_t. Nevertheless, with the knowledge of v_t, we can further improve the regret bound under the additional smoothness assumption (Assumption 5.2). In particular, the discretization regret in episode k is reduced to $\mathcal{O}\left(T_k/N_k^2\right)$.

Corollary 5.4. Under Assumption 2.1, Assumption 5.2, and Assumption 5.1, the expected regret of Algorithm 4 satisfies

$$\mathbb{E}[\mathrm{Reg}(T)] = \mathcal{O}\left(T^{\frac{3}{5}}\ln^{\frac{2}{5}}T + \sqrt{\rho_{\mathcal{V}}(\delta)T}\right)$$

with probability at least $1-\delta$.

This bound matches the lower bound $\Omega\left(T^{\frac{3}{5}}\right)$ reported in Luo et al. (2022, 2024). However, since the models in Luo et al. (2022, 2024) rely solely on censored demand data (i.e., observing y_t rather than v_t), it remains unclear whether sellers can achieve a lower regret when full valuation information is available.

Nevertheless, in Section 5.3 we show that this regret lower bound is attainable using only the purchasing decision y_t, provided that a powerful classification oracle is available.

5.3. Classification oracle

To estimate the unknown value function v^*, we introduce a method that utilizes all the data (x_t, p_t, y_t) from the previous $(k-1)$-th episode. This yields an estimate $\hat{v}_k$ that can be used for pricing in the k-th episode. Since y_t is binary and determined by x_t and p_t via $\mathbb{P}(y_t=1) = 1-F(p_t - v^*(x_t))$, we have

$$\mathbb{P}(y_t=1)\begin{cases} > \frac{1}{2}, & \text{if } F^{-1}\left(\frac{1}{2}\right) + v^*(x_t) - p_t > 0, \\ = \frac{1}{2}, & \text{if } F^{-1}\left(\frac{1}{2}\right) + v^*(x_t) - p_t = 0, \\ < \frac{1}{2}, & \text{if } F^{-1}\left(\frac{1}{2}\right) + v^*(x_t) - p_t < 0. \end{cases}$$

This observation converts the estimation problem into a classification problem. By applying a classification oracle, we can obtain a predictor $\hat{v} \in \mathcal{V}$ that approximates v^*. The classification oracle effectively provides a decision boundary based on the unknown function v^*.

Assumption 5.3. Given n samples of the form (x_i, p_i, y_i), the classification oracle returns a predictor $\hat{v} \in \mathcal{V}$. For any $\delta > 0$, with probability at least $1-\delta$, we have

$$\mathbb{E}_x[(\hat{v}(x) - v^*(x))^2] \le \frac{\rho_{\mathcal{V}}(\delta)}{n^\alpha}$$

for some positive α.

In the work of Luo et al. (2024), a linear classification method is applied to the adaptive data from the previous episode to obtain a linear parameter estimator. They demonstrate that in the linear valuation model, Assumption 5.3 is satisfied with $\alpha \geq 1$. However, for a general function space, it remains unclear whether one can design a suitable classification oracle that meets the required conditions. Typically, classification oracles are criticized for the lack of theoretical guarantees regarding their estimation accuracy. As Luo et al. (2022) admits, "α is indeterministic and no rigorous justification has been made."

Under the Lipschitz condition alone, the upper bound for the contextual dynamic pricing problem is $\tilde{\mathcal{O}}(\rho_{\mathcal{V}}(\delta)^{\frac{1}{3}}T^{\frac{2}{3}})$. However, by incorporating the additional assumption of second-order smoothness (Assumption 5.2), we can enhance the performance relative to previous work (Luo et al. 2024).

Corollary 5.5. Under Assumption 2.1, Assumption 5.2 and Assumption 5.3, the expected regret of modifed Algorithm 5 satisfies

$$\mathbb{E}[\mathrm{Reg}(T)] = \mathcal{O}(T^{\frac{3}{5}}\ln^{\frac{2}{5}}T + \sqrt{\rho_{\mathcal{V}}(\delta)}T^{1-\frac{\alpha}{2}}).$$

with probability at least $1-\delta$.

It is worth noting that to achieve minimax optimality, it only requires $\alpha \geq \frac{4}{5}$, which is slightly less than 1. If we specify $\mathcal{V}$ as a linear function space and employ logistic regression to estimate the unknown parameters, we can improve upon the results in Luo et al. (2024) and match the lower bound $\Omega\left(T^{\frac{3}{5}}\right)$ reported in Luo et al. (2022).

6. Numerical Experiments

In this section, we present numerical simulations to evaluate the empirical performance of our algorithm for linear valuation models.

6.1. Comparison with existing methods

We consider the contextual dynamic pricing problem under randomly generated linear valuation models. The contexts are drawn from a standard Gaussian distribution in $\mathbb{R}^4$, followed by ℓ_2-normalization to a unit ball. The noise term is drawn from a Gaussian distribution $\mathcal{N}(0, 0.3)$, truncated between -1 and 1. Similarly, the parameter θ is initialized as a normalized vector, with its components drawn from a standard Gaussian distribution.

We compare our method with the algorithms proposed in Tullii et al. (2024) and Fan et al. (2024). We note that the algorithm in Tullii et al. (2024) requires knowledge of the Lipschitz constant L in Assumption 2.2. For consistency with their experimental setup, we adopt $L = 1$ for their method

while the true Lipschitz constant is approximately equal to 0.78. For the both Tullii et al. (2024) and Fan et al. (2024), we use the implementation release publicly by Tullii et al. (2024). We evaluate all methods under the price bound $B = 2$ for all time horizons $T \in \{1000, 2000, 5000, 10000, 50000\}$. The results are averaged across 10 independently generated instances.

The average cumulative regret achieved by the three algorithms is shown in Figure 1, together with the 95% confidence intervals for the mean regret. Our algorithm consistently demonstrates superior cumulative regret (up to $80-90\%$ lower) compared to both benchmarks across all time horizons. Notably, the variance across 10 runs, as indicated by the confidence intervals, is significantly smaller than that of Tullii et al. (2024) and Fan et al. (2024), reflecting enhanced stability. This improvement is attributed to our adaptive layer search mechanism, which dynamically optimizes upper confidence bounds without requiring prior knowledge of the Lipschitz constant.

While Tullii et al. (2024) achieves faster runtime through Lipschitz-constant-guided UCB simplification, our method eliminates this dependency through layer-wise UCB at only 1.8-4.5 times the computation time of Tullii et al. (2024). This is particularly impactful in real-world pricing systems where Lipschitz constants are rarely known a priori. Furthermore, we significantly outperform Fan et al. (2024) in speed by avoiding their computationally intensive full distribution estimation.

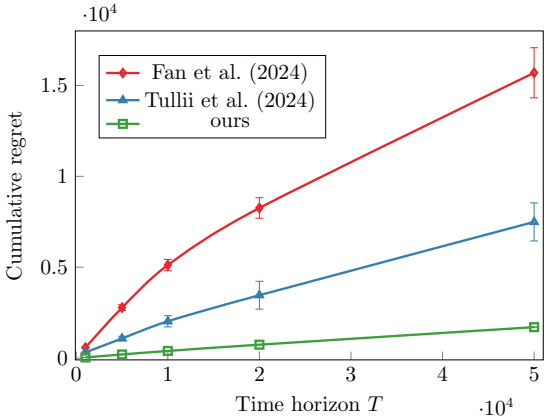


Figure 1 Comparison of the expected cumulative regret of our algorithm with Fan et al. (2024), Tullii et al. (2024). The results are averaged over 10 independent random instances, with 95% confidence intervals shown.

6.2. Noise distribution

In this experiment, we investigate the impact of the noise distribution on the performance of our algorithm. We conduct simulations using noise distributions with varying smoothness properties, namely the standard normal, the standard Cauchy, and the uniform distributions. The CDF of

the standard normal distribution is infinitely differentiable and possesses moments of all orders. Although the Cauchy distribution is also infinitely differentiable, it does not have a well-defined mean. In contrast, the CDF of the uniform distribution is not differentiable.

For each noise distribution, we again generate random context vectors of dimension $d = 4$ and set the price bound $B = 2$, as in Section 6.1. The noise distributions are all truncated to the same interval of $[-1, 1]$. We test our algorithm over time horizons $T \in \{1000, 2000, 4000, 8000\}$. The results are reported in Figure 2, which is averaged over 10 independent replications with 95% confidence intervals presented.

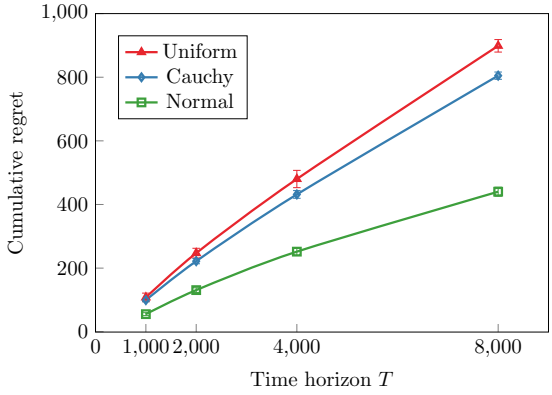


Figure 2 Comparison of the expected cumulative regret under various noise distributions. The results are averaged over 10 independent random instances, with 95% confidence intervals shown.

We observe that the smoothness of the noise distribution has a significant impact on performance, with smoother distribution leading to lower regret. This is consistent with the observations from Javanmard and Nazerzadeh (2019), where the regret rate can be reduced to $\tilde{\mathcal{O}}(\sqrt{T})$ if the noise distribution belongs to the exponential family. This result is also confirmed theoretically by Fan et al. (2024), Chen et al. (2024). However, as the smoothness of the noise distribution decreases, we see a noticeable increase in regret. We remark that our algorithm is agnostic to the noise distribution, as along as the basic Lipschitz continuity Assumption 2.2 holds.

7. Conclusion

We offers a comprehensive solution to the contextual dynamic pricing problem with minimax optimality and practical applicability. Our algorithm integrates an explore-then-UCB strategy with layered data partitioning to yield a regret upper bound of $\tilde{\mathcal{O}}(\rho_{\mathcal{V}}^{\frac{1}{3}}(\delta)T^{\frac{2}{3}})$. It achieves improved regret bounds compared to existing approaches for linear valuation models and extends to more

general function spaces via suitable offline regression oracles while relying on minimal smoothness assumptions on the noise distribution.

Future research can pursue several directions. First, relaxing the Lipschitz continuity assumption on the noise distribution F and investigating the corresponding regret lower bound would deepen our understanding of the problem's intrinsic difficulty. Since Xu and Wang (2022) provide a regret upper bound of $\tilde{\mathcal{O}}(T^{\frac{3}{4}})$ without Lipschitz continuity, it is conjectured that the lower bound remains $\Omega(T^{\frac{3}{4}})$ when this assumption is removed.

Second, further investigating the constant factor in the regret bound with respect to the valuation function space complexity is important. For linear valuation models with covariates of dimension d_0, we established a regret bound of $\tilde{\mathcal{O}}(d_0^{\frac{1}{3}}T^{\frac{2}{3}})$. It remains an open question whether this bound is optimal with respect to the covariate dimension d_0. Addressing this issue would reveal the dependence of regret on the complexity of the underlying valuation function space.

Finally, improving regret upper bounds for smoother distributions is a promising direction. By incorporating additional second-order smoothness conditions, the regret upper bound may be improved to $\tilde{\mathcal{O}}(T^{\frac{3}{5}})$. Further investigation in this area would clarify the impact of distribution smoothness on regret performance.

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Appendices

Appendix A Algorithms

Algorithm 3 Distribution-Dependent Dynamic Pricing Algorithm

Input: price upper bound B, distribution F, confidence parameter δ

for episode $k = 1, 2, \cdots$ **do**

  Feed the collected data in the previous episode to the **General Offline Regression Oracle** and obtain an estimator $\hat{v}_k$ (For $k = 1$, set $\hat{v}_k = 0$)

  for round $t = 2^{k-1} + 1, \cdots, 2^k$ **do**

    Observe context x_t

    Set a price $p_t = \arg\max_p p(1 - F(p - \hat{v}_k(x_t)))$

    Observe the feedback y_t

Algorithm 4 Distribution-Free Dynamic Pricing Algorithm with General Offline Regression Oracle and Observable Valuation

Input: price upper bound B, estimation parameter $\rho_{\mathcal{V}}(\delta)$

for episode $k = 1, 2, \cdots$ **do**

  Feed the collected data in the previous episode to the **General Offline Regression Oracle** and obtain an estimator $\hat{v}_k$

  Set the length $T_k = 2^{k-1}$

  Set the discretization number $N_k = \lceil T_k^{\frac{1}{5}} \rceil$

  for $t = 2^{k-1} + 1, \cdots, 2^k$ **do**

    Apply the **UCB-LDP** on the coming sequential contexts x_t with the estimator $\hat{v}_k$, the discretization number N_k, the length T_k and the bound B

Appendix B Concentration Inequality

Lemma B.1. (Azuma's Inequality) Let $X_1, \cdots, X_n$ be a martingale difference sequence with respect to filtration $\{\mathcal{F}_t\}_{1 \le t \le n}$. Suppose that $|X_\tau| \le a_\tau$ for some $a_\tau > 0, \tau \in [n]$. Then

$$\mathbb{P}\left(\left| \frac{1}{n} \sum_{\tau=1}^{n} X_\tau \right| \ge 2\sqrt{\ln(2/\delta) \sum_{\tau=1}^{n} a_\tau^2} \right) \le \delta.$$

Algorithm 5 Distribution-Free Dynamic Pricing Algorithm with General Offline Regression Oracle

Input: price upper bound B, estimation parameter $\rho_{\mathcal{V}}(\delta)$

Let $k^* = \lceil 3\log_2 \rho_{\mathcal{V}}(\delta) \rceil$

for round $t = 1, 2, \cdots, 2^{k^*-1}$ **do** ▷ Warm up

  Observe context x_t

  Set a price p_t uniformly from $(0, B)$

  Observe the feedback y_t

for episode $k = k^*, k^*+1, \cdots$ **do**

  Set the length of the k-th episode as $\ell_k = 2^{k-1}$

  Set the length of the exploration phase at k-th episode as $T_k^e = \lceil \ell_k^{\frac{2}{3}} \rho_{\mathcal{V}}(\delta) \rceil$

  for round $t = 2^{k-1}+1, \cdots, 2^{k-1}+T_k^e$ **do** ▷ Exploration phase

    Observe context x_t

    Set a price p_t uniformly from $(0, B)$

    Observe the feedback y_t

  Find the estimator $\hat{v}_k$ by the **General Offline Regression Oracle** ▷ Estimation phase

  Set the length $T_k = \ell_k - T_k^e$

  Set the discretization number $N_k = \lceil T_k^{\frac{1}{3}} \rceil$

  for $t = 2^{k-1}+T_k^e+1, \cdots, 2^k$ **do** ▷ UCB phase

    Apply the **UCB-LDP** on the coming sequential contexts x_t with the estimator $\hat{v}_k$, the discretization number N_k, the length T_k and the bound B

Appendix C Proofs

C.1 Proofs for Section 3

Lemma 3.1. With probability $1-\delta$, the estimator $\hat{v}_k$ in Algorithm 1 satisfies

$$\mathbb{E}[(\hat{v}_k(x) - v^*(x))^2] \le \frac{18B\ln(|\mathcal{V}|\log_2(T_k^e)/\delta)}{T_k^e}.$$

Proof. Consider the data collected at the episode k. Then for any round t satisfying $\ell_k < t \le \ell_k + T_k^e$ and any $v \in \mathcal{V}$, we denote the random variable

$$Z_t = (v(x_t) - By_t)^2 - (v^*(x_t) - By_t)^2.$$

Furthermore, we have

$$\mathbb{E}[Z_t] = \mathbb{E}[(v_k(x_t) - By_t)^2 - (By_t - v^*(x_t))^2]$$

$$
\begin{aligned}
&= \mathbb{E}[(v(x_t) - v^*(x_t))(v(x_t) + v^*(x_t) - 2By_t)] \\
&= \mathbb{E}[(v(x_t) - v^*(x_t))\mathbb{E}_{y_t}[(v(x_t) + v^*(x_t) - 2By_t)|x_t]] \\
&= \mathbb{E}[(v(x_t) - v^*(x_t))^2].
\end{aligned}
$$

Therefore, we obtain

$$
\begin{aligned}
\mathrm{Var}(Z_t) \le \mathbb{E}[Z_t^2] &= \mathbb{E}[(v(x_t) - v^*(x_t))^2(v(x_t) + v^*(x_t) - 2By_t)^2] \\
&\le 4B\mathbb{E}[(v(x_t) - v^*(x_t))^2] \\
&= 16B^2\mathbb{E}[Z_t].
\end{aligned}
$$

From Freedman's inequality, we have

$$
\begin{aligned}
\sum_{t=\ell_k+1}^{\ell_k+T_k^e} \mathbb{E}[Z_t] - \sum_{t=\ell_k+1}^{\ell_k+T_k^e} Z_t &\le 4\sqrt{\sum_{t=\ell_k+1}^{\ell_k+T_k^e} \mathrm{Var}(Z_t)\ln(1/\delta)} + 2B\ln(1/\delta) \\
&\le 16B\sqrt{\sum_{t=\ell_k+1}^{\ell_k+T_k^e} \mathbb{E}[Z_t]\ln(1/\delta)} + 2B\ln(1/\delta).
\end{aligned}
$$

with probability at least $1 - \delta\log_2(T_k^e)$. The last inequality is due to the boundedness of y_t. Solving $\sum_{t=\ell_k+1}^{\ell_k+T_k^e} \mathbb{E}[Z_t]$ from the above inequality yields

$$
\sum_{t=\ell_k+1}^{\ell_k+T_k^e} \mathbb{E}[Z_t] \le 18B\ln(1/\delta) + \sum_{t=\ell_k+1}^{\ell_k+T_k^e} Z_t.
$$

By a union bound, with probability at least $1 - |\mathcal{V}|\log_2(T_k^e)\delta$,

$$
\mathbb{E}[(v(x_t) - v^*(x_t))^2] \le \frac{18B\ln(1/\delta)}{T_k^e} + \frac{1}{T_k^e}\sum_{t=\ell_k+1}^{\ell_k+T_k^e} Z_t.
$$

Then from the algorithmic construction, we know $\sum_{t=\ell_k+1}^{\ell_k+T_k^e} Z_t \le 0$ for $\hat{v}_k$. We immediately obtain that

$$
\mathbb{E}[(\hat{v}_k(x_t) - v^*(x_t))^2] \le \frac{18B\ln(1/\delta)}{T_k^e}
$$

holds with probability at least $1 - |\mathcal{V}|\log_2(T_k^e)\delta$. □

Lemma 3.2. Fix an episode k. For any round t in the episode k, and any layer s at round t, with probability at least $1 - \frac{\delta}{N_k T_k}$, each $j \in [N_k]$ satisfies

$$
|\xi_j^* - w_{t,s}^j| \le r_{t,s}^j + L\eta_{t,s}^j, \quad \text{where} \quad \eta_{t,s}^j = \frac{1}{|\Psi_t^s(j)|}\sum_{\tau \in \Psi_t^s(j)} |\hat{v}_k(x_\tau) - v^*(x_\tau)|
$$

Proof. Let $\mathcal{H}_0$ denote the filtration generated by

$$\left\{(\tau, x_\tau, p_\tau), \tau \in \bigcup_{s' \le s} \Psi_t^{s'}\right\} \cup \left\{y_\tau, \tau \in \bigcup_{s' < s} \Psi_t^{s'}\right\}.$$

Based on the algorithmic construction, the selection of round τ into Ψ_t^s depends solely on $\mathcal{H}_0$. Therefore, condition on $\mathcal{H}_0$, $\{\epsilon_\tau : \tau \in \Psi_t^s\}$ are independent random variables. Considering the conditional independence of ϵ_τ indexed in Ψ_t^s, we can deduce that for $\tau \in \Psi_t^s(j)$

$$\begin{aligned}
\mathbb{E}[y_\tau | \{y_{\tau'} : \tau' \in \Psi_t^s(j), \tau' < \tau\}, \mathcal{H}_0] &= \mathbb{E}[\mathbf{I}_{\{\epsilon_\tau > m_j + \hat{v}_k(x_\tau) - v^*(x_\tau)\}} | \{\epsilon_{\tau'} : \tau' \in \Psi_t^s(j), \tau' < \tau\}, \mathcal{H}_0] \\
&= \mathbb{E}[\mathbf{I}_{\{\epsilon_\tau > m_j + \hat{v}_k(x_\tau) - v^*(x_\tau)\}} | \mathcal{H}_0] \\
&= 1 - F(m_j + \hat{v}_k(x_\tau) - v^*(x_\tau)).
\end{aligned}$$

Now we consider the concentration for $y_\tau, \tau \in \Psi_t^s(j)$. For notation brevity, denote $\mathcal{H}_\tau$ as the filtration generated by $\{y_{\tau'} : \tau' \in \Psi_t^s(j), \tau' < \tau\} \cup \mathcal{H}_0$. Then for $\tau = 0$, the definition of $\mathcal{H}_\tau$ is consistent with $\mathcal{H}_0$. To apply Azuma's inequality, it is easy to check $y_\tau - \mathbb{E}[y_\tau | \mathcal{H}_\tau]$ is a martingale difference sequence adapted to filtration $\mathcal{H}_\tau$. Hence, we can derive the following results:

$$\mathbb{P}\left(\left| w_{t,s}^j - \frac{\sum_{\tau \in \Psi_t^s(j)} \mathbb{E}[y_\tau | \mathcal{H}_\tau]}{|\Psi_t^s(j)|} \right| \ge \sqrt{\frac{\ln(N_k T_k / \delta)}{4 |\Psi_t^s(j)|}}\right) \le \frac{\delta}{N_k T_k}.$$

Hence, we obtain

$$\mathbb{P}\left(\left| w_{t,s}^j - \frac{\sum_{\tau \in \Psi_t^s(j)} \mathbb{E}[y_\tau | \mathcal{H}_\tau]}{|\Psi_t^s(j)|} \right| \ge r_{t,s}^j\right) \le \mathbb{P}\left(\left| w_{t,s}^j - \frac{\sum_{\tau \in \Psi_t^s(j)} \mathbb{E}[y_\tau | \mathcal{H}_\tau]}{|\Psi_t^s(j)|} \right| \ge \sqrt{\frac{\ln(N_k T_k / \delta)}{4 |\Psi_t^s(j)|}}\right) \le \frac{\delta}{N_k T_k}.$$

From Assumption 2.2, we can obtain

$$\left| \frac{\sum_{\tau \in \Psi_t^s(j)} (\xi_j^* - \mathbb{E}[y_\tau | \mathcal{H}_\tau])}{|\Psi_t^s(j)|} \right| \le \frac{L \sum_{\tau \in \Psi_t^s(j)} |\hat{v}_k(x_\tau) - v^*(x_\tau)|}{|\Psi_t^s(j)|},$$

so by the triangular inequality, we find

$$\begin{aligned}
&\mathbb{P}\left(\left| w_{t,s}^j - \xi_j^* \right| \ge r_{t,s}^j + \frac{L \sum_{\tau \in \Psi_t^s(j)} |\hat{v}_k(x_\tau) - v^*(x_\tau)|}{|\Psi_t^s(j)|}\right) \\
\le &\mathbb{P}\left(\left| w_{t,s}^j - \frac{\sum_{\tau \in \Psi_t^s(j)} \mathbb{E}[y_\tau | \mathcal{H}_\tau]}{|\Psi_t^s(j)|} \right| + \left| \frac{\sum_{\tau \in \Psi_t^s(j)} (\xi_j^* - \mathbb{E}[y_\tau | \mathcal{H}_\tau])}{|\Psi_t^s(j)|} \right| \right. \\
&\qquad \left. \ge r_{t,s}^j + \frac{L \sum_{\tau \in \Psi_t^s(j)} |\hat{v}_k(x_\tau) - v^*(x_\tau)|}{|\Psi_t^s(j)|}\right) \\
\le &\mathbb{P}\left(\left| w_{t,s}^j - \frac{\sum_{\tau \in \Psi_t^s(j)} \mathbb{E}[y_\tau | \mathcal{H}_\tau]}{|\Psi_t^s(j)|} \right| \ge r_{t,s}^j\right) \\
\le &\frac{\delta}{N_k T_k}.
\end{aligned}$$

□

C.2 Proofs for upper bounds in Section 4.1

Lemma 4.1. For each round t in episode k and each layer $s \in [s_t - 1]$, under the high probability event Γ_k, we have

$$\mathbb{E}_{x_t}\left[\mathsf{Rev}_t\left(\tilde{p}_t^*\right) - \mathsf{Rev}_t\left(\tilde{p}_{t,s}^*\right)\right] \le 4BL\mathbb{E}_x\left[\left|\hat{v}_k(x) - v^*(x)\right|\right](s-1).$$

Proof. We prove this lemma by induction on s. For $s = 1$, the lemma holds naturally since $\mathcal{A}_{t,1} = [N_k]$ and hence $j_t^* = j_{t,1}^*$. Assume that the bound holds at layer s. It suffices to show that

$$\mathbb{E}_{x_t}\left[\mathsf{Rev}_t\left(m_{j_{t,s}^*} + \hat{v}_k(x_t)\right) - \mathsf{Rev}_t\left(m_{j_{t,s+1}^*} + \hat{v}_k(x_t)\right)\right] \le 4BL\mathbb{E}_x\left[\left|\hat{v}_k(x) - v^*(x)\right|\right].$$

If $j_{t,s}^* = j_{t,s+1}^*$, then the desired bound holds. Hence we assume that $j_{t,s}^* \notin \mathcal{A}_{t,s+1}$. Let

$$\hat{j}_{t,s} := \arg\max_{j \in \mathcal{A}_{t,s}} \left(m_j + \hat{v}_k(x_t)\right)\left(w_{t,s}^j + r_{t,s}^j\right)$$

be the index with the highest UCB in $\mathcal{A}_{t,s}$. From Step 9 of Algorithm 2, we know that $\hat{j}_{t,s} \in \mathcal{A}_{t,s+1}$. Then we have

$$\begin{aligned}
&\mathsf{Rev}_t\left(m_{j_{t,s}^*} + \hat{v}_k(x_t)\right) - \mathsf{Rev}_t\left(m_{j_{t,s+1}^*} + \hat{v}_k(x_t)\right) \\
\le\ &\mathsf{Rev}_t\left(m_{j_{t,s}^*} + \hat{v}_k(x_t)\right) - \mathsf{Rev}_t\left(m_{\hat{j}_{t,s}} + \hat{v}_k(x_t)\right) \\
\le\ &\left(m_{j_{t,s}^*} + \hat{v}_k(x_t)\right)\left(1 - F\left(m_{j_{t,s}^*}\right)\right) - \left(m_{\hat{j}_{t,s}} + \hat{v}_k(x_t)\right)\left(1 - F\left(m_{\hat{j}_{t,s}}\right)\right) + 2BL\left|\hat{v}_k(x_t) - v^*(x_t)\right|.
\end{aligned}$$

From the definition to Γ_k, we know that

$$\begin{aligned}
&\left(m_{j_{t,s}^*} + \hat{v}_k(x_t)\right)\left(1 - F\left(m_{j_{t,s}^*}\right)\right) - \left(m_{\hat{j}_{t,s}} + \hat{v}_k(x_t)\right)\left(1 - F\left(m_{\hat{j}_{t,s}}\right)\right) \\
\le\ &\left(m_{j_{t,s}^*} + \hat{v}_k(x_t)\right)\left(w_{t,s}^{j_{t,s}^*} + r_{t,s}^{j_{t,s}^*}\right) - \left(m_{\hat{j}_{t,s}} + \hat{v}_k(x_t)\right)\left(w_{t,s}^{\hat{j}_{t,s}} - r_{t,s}^{\hat{j}_{t,s}}\right) + BL\eta_{t,s}^{j_{t,s}^*} + BL\eta_{t,s}^{\hat{j}_{t,s}} \\
=\ &\left(m_{j_{t,s}^*} + \hat{v}_k(x_t)\right)\left(w_{t,s}^{j_{t,s}^*} + r_{t,s}^{j_{t,s}^*}\right) - \left(m_{\hat{j}_{t,s}} + \hat{v}_k(x_t)\right)\left(w_{t,s}^{\hat{j}_{t,s}} + r_{t,s}^{\hat{j}_{t,s}}\right) + 2\left(m_{\hat{j}_{t,s}} + \hat{v}_k(x_t)\right) r_{t,s}^{\hat{j}_{t,s}} \\
&+ BL\eta_{t,s}^{j_{t,s}^*} + BL\eta_{t,s}^{\hat{j}_{t,s}}.
\end{aligned}$$

From the algorithmic construction (Step 12), we know

$$\left(m_{\hat{j}_{t,s}} + \hat{v}_k(x_t)\right) r_{t,s}^{\hat{j}_{t,s}} \le B2^{-s}$$

as $s < s_t$. Since $j_{t,s}^* \notin \mathcal{A}_{t,s+1}$, Step 9 yields

$$\left(m_{j_{t,s}^*} + \hat{v}_k(x_t)\right)\left(w_{t,s}^{j_{t,s}^*} + r_{t,s}^{j_{t,s}^*}\right) - \left(m_{\hat{j}_{t,s}} + \hat{v}_k(x_t)\right)\left(w_{t,s}^{\hat{j}_{t,s}} + r_{t,s}^{\hat{j}_{t,s}}\right) < -B2^{1-s} \le -2\left(m_{\hat{j}_{t,s}} + \hat{v}_k(x_t)\right) r_{t,s}^{\hat{j}_{t,s}}.$$

Combining all the inequalities above, we obtain

$$\mathsf{Rev}_t\left(m_{j_{t,s}^*} + \hat{v}_k(x_t)\right) - \mathsf{Rev}_t\left(m_{j_{t,s+1}^*} + \hat{v}_k(x_t)\right) \le 2BL\left|\hat{v}_k(x_t) - v^*(x_t)\right| + BL\eta_{t,s}^{j_{t,s}^*} + BL\eta_{t,s}^{\hat{j}_{t,s}}.$$

Recall that $\mathbb{E}_x[\eta_{t,s}^j] = \mathbb{E}_x[|\hat{v}_k(x) - v^*(x)|]$ for any t, s, j, taking expectation with respect to x_τ yields the desired inequality. □

Lemma 4.2. For each round t in episode k and each layer $2 \le s \le s_t - 1$, under the high probability event Γ_k, we have

$$\mathbb{E}_{x_t}\left[\mathsf{Rev}_t\left(\tilde{p}^*_{t,s}\right) - \mathsf{Rev}_t\left(p_t\right)\right] \le 8B \cdot 2^{-s} + 4BL\mathbb{E}_x\left[|\hat{v}_k(x) - v^*(x)|\right].$$

Proof. For all $2 \le s \le s_t$, Step 9 shows that

$$(m_j + \hat{v}_k(x_t))(w^j_{t,s-1} + r^j_{t,s-1}) \ge (m_{j^*_{t,s}} + \hat{v}_k(x_t))(w^{j^*_{t,s}}_{t,s-1} + r^{j^*_{t,s}}_{t,s-1}) - B2^{2-s}, \forall j \in \mathcal{A}_{t,s},$$

as $j^*_{t,s} \in \mathcal{A}_{t,s} \subset \mathcal{A}_{t,s-1}$. Step 12 implies that $(m_j + \hat{v}_k(x_t))r^j_{t,s-1} \le B2^{1-s}$ for all $j \in \mathcal{A}_{t,s-1}$. Combing two inequalities we obtain

$$\begin{aligned}
&(m_{j^*_{t,s}} + \hat{v}_k(x_t))(w^{j^*_{t,s}}_{t,s-1} + r^{j^*_{t,s}}_{t,s-1}) - (m_j + \hat{v}_k(x_t))(w^j_{t,s-1} - r^j_{t,s-1}) \\
\le& 2(m_j + \hat{v}_k(x_t))r^j_{t,s-1} + B2^{2-s} \\
\le& 4B2^{1-s}, \forall j \in \mathcal{A}_{t,s}.
\end{aligned}$$

Therefore, from the definition of Γ_k we have

$$\begin{aligned}
4B2^{1-s} &\ge (m_{j^*_{t,s}} + \hat{v}_k(x_t))(w^{j^*_{t,s}}_{t,s-1} + r^{j^*_{t,s}}_{t,s-1}) - (m_j + \hat{v}_k(x_t))(w^j_{t,s-1} - r^j_{t,s-1}) \\
&\ge \mathbb{E}_{x_t}\left[\mathsf{Rev}_t\left(\tilde{p}^*_{t,s}\right) - \mathsf{Rev}_t\left(p_t\right)\right] - 2BL\eta^{j^*_{t,s}}_{t,s-1} - 2BL\eta^{j_{t,s}}_{t,s-1}.
\end{aligned}$$

Recall that $\mathbb{E}_x[\eta^j_{t,s}] = \mathbb{E}_x[|\hat{v}_k(x) - v^*(x)|]$ for any t, s, j, taking expectation with respect to x_τ yields the desired inequality. □

Lemma C.1. Let Ψ_0 be the set of rounds for which an alternative is chosen when $(m_j + \hat{v}_k(x_t))r^j_{t,s} \le 1/\sqrt{T_k}$ for all $j \in \mathcal{A}_{t,s}$. Given the event Γ_k, for every round $t \in \Psi_0$ in the episode k, we have

$$\mathbb{E}_{x_t}[\mathsf{Rev}_t(\tilde{p}^*_t) - \mathsf{Rev}_t(p_t)] \le \frac{2B}{\sqrt{T_k}} + 4BL\mathbb{E}_x[|\hat{v}_k(x) - v^*(x)|].$$

Proof. It follows from Assumption 2.2 and Lemma 4.2 that

$$\begin{aligned}
&\mathbb{E}_{x_t}[\mathsf{Rev}_t(\tilde{p}^*_t) - \mathsf{Rev}_t(p_t)] \\
=&\mathbb{E}_{x_t}[\mathsf{Rev}_t(m_{j^*_t} + \hat{v}_k(x_t)) - \mathsf{Rev}_t(m_j + \hat{v}_k(x_t))] \\
\le&\mathbb{E}_{x_t}[(m_{j^*_t} + \hat{v}_k(x_t))(1 - F(m_{j^*_t})) - (m_{j_t} + \hat{v}_k(x_t))(1 - F(m_{j_t}))] + 2BL\mathbb{E}_x[|\hat{v}_k(x) - v^*(x_k)|] \\
\le&\mathbb{E}_{x_t}[(m_{j^*_t} + \hat{v}_k(x_t))(w^{j^*_t}_{t,1} + r^{j^*_t}_{t,1}) - (m_{j_t} + \hat{v}_k(x_t))(w^{j_t}_{t,1} - r^{j_t}_{t,1})] + 4BL\mathbb{E}_x[|\hat{v}_k(x) - v^*(x)|] \\
\le&\frac{2B}{\sqrt{T_k}} + \mathbb{E}_{x_t}[(m_{j^*_t} + \hat{v}_k(x_t))(w^{j^*_t}_{t,1} + r^{j^*_t}_{t,1}) - (m_{j_t} + \hat{v}_k(x_t))(w^{j_t}_{t,1} + r^{j_t}_{t,1})] + 4BL\mathbb{E}_x[|\hat{v}_k(x) - v^*(x)|] \\
\le&\frac{2B}{\sqrt{T_k}} + 4BL\mathbb{E}_x[|\hat{v}_k(x) - v^*(x)|].
\end{aligned}$$

□

Lemma C.2. Assuming $|\Psi^s_{T_k+1}| \geq 2$, we have

$$\sum_{t\in\Psi^s_{T_k+1}} p_t r^{j_t}_{t,s} \leq B\sqrt{2N_k|\Psi^s_{T_k+1}|\ln(N_kT_k/\delta)}.$$

Proof. We have

$$\sum_{t\in\Psi^s_{T_k+1}} p_t r^{j_t}_{t,s} = \sqrt{\ln(N_kT_k/\delta)}\sum_{t\in\Psi^s_{T_k+1}} \frac{p_t}{\sqrt{2|\Psi^s_t(j_t)|}}.$$

It suffices to bound the term $\sum_{t\in\Psi^s_{T_k+1}} \frac{p_t}{\sqrt{2|\Psi^s_t(j_t)|}}$. Notice that

$$\begin{aligned}
\sum_{t\in\Psi^s_{T_k+1}} \frac{p_t}{\sqrt{2|\Psi^s_t(j_t)|}} &\leq \sum_{j=1}^{N_k}\sum_{t=1}^{|\Psi^s_{T_k}(j)|} \frac{B}{\sqrt{2t}} \\
&\leq \sum_{j=1}^{N_k} B\sqrt{2|\Psi^s_{T_k}(j)|} \\
&\leq B\sqrt{2N_k\sum_{j=1}^{N_k}|\Psi^s_{T_k}(j)|} \\
&\leq B\sqrt{2N_k|\Psi^s_{T_k+1}|}.
\end{aligned}$$

Therefore, we conclude that

$$\sum_{t\in\Psi^s_{T_k+1}} p_t r^{j_t}_{t,s} \leq B\sqrt{2N_k|\Psi^s_{T_k+1}|\ln(N_kT_k/\delta)}.$$

□

Lemma C.3. For all s, we have

$$|\Psi^s_{T_k+1}| \leq 2^s\sqrt{2N_k|\Psi^s_{T_k+1}|\ln(N_kT_k/\delta)}.$$

Proof. From Step 15, we know that

$$\sum_{t\in\Psi^s_{T_k+1}} p_t r^{j_t}_{t,s} \geq B2^{-s}|\Psi^s_{T_k+1}|.$$

By Lemma C.2, we obtain

$$\sum_{t\in\Psi^s_{T_k+1}} p_t r^{j_t}_{t,s} \leq B\sqrt{2N_k|\Psi^s_{T_k+1}|\ln(N_kT_k/\delta)}.$$

Therefore, combing above inequalities, we have

$$|\Psi^s_{T_k+1}| \leq 2^s\sqrt{2N_k|\Psi^s_{T_k+1}|\ln(N_kT_k/\delta)}.$$

□

Lemma 4.4. Under Assumption 2.1 and Assumption 2.2, the expected learning regret $\mathbb{E}\left[\sum_{t=1}^{T}\mathcal{R}_t^1\right]$ is upper bounded by

$$\sum_{k=1}^{\lceil\log_2 T\rceil}\left(8B\sqrt{2N_kT_k\ln(T_kN_k/\delta)\ln T_k}+4BL\ln^{\frac{1}{3}}(|\mathcal{V}|\ln T/\delta)(k\ell_k)^{\frac{2}{3}}+2B\sqrt{T_k}+8N_k\ln(T_kN_k/\delta)\right)$$

with probability at least $1-\delta$.

Proof. We consider the episode k. Let Ψ_0 be the set of rounds for which an alternative is chosen when $(m_j+\hat{v}_k(x_t))r_{t,s}^j \le B/\sqrt{T_k}$. Since $2^{-S}\le 1/\sqrt{T_k}$, we have $\Psi_0\cup\left(\bigcup_{s\in[S]}\Psi_{T_k+1}^s\right)=[T_k]$. Thanks to Lemma C.1, we have

$$\mathbb{E}\left[\sum_{t\in\Psi_0}\mathcal{R}_t^1\right]\le 2B\sqrt{T_k}+2BL\mathbb{E}_x[|\hat{v}_k(x)-v^*(x)|]T_k.$$

Recall $j_t^*:=\arg\max_{j\in[N_k]}(m_j+\hat{v}_k(x_t))(1-F(m_j+\hat{v}_k(x_t)-v^*(x_t)))$ and we define the discrete empirical best price as $\tilde{p}_t^*=m_{j_t^*}+\hat{v}_k(x_t)$. We first bound the regret $\mathbb{E}\left[\sum_{t=2^{k-1}+T_k^e+1}^{2^k}\mathcal{R}_t^1\right]$ at the episode k as following:

$$\begin{aligned}
\mathbb{E}\left[\sum_{t=2^{k-1}+T_k^e+1}^{2^k}\mathcal{R}_t^1\right] &= \sum_{t=2^{k-1}+T_k^e+1}^{2^k}\mathbb{E}\Big[\tilde{p}_t^*\Big(1-F\big(\tilde{p}_t^*-v^*(x_t)\big)\Big)-p_t\Big(1-F\big(p_t-v^*(x_t)\big)\Big)\Big]\\
&= \sum_{t=2^{k-1}+T_k^e+1}^{2^k}\mathbb{E}\Big[\mathsf{Rev}_t\big(\tilde{p}_t^*\big)-\mathsf{Rev}_t\big(p_t\big)\Big]\\
&= \sum_{s=1}^{S}\sum_{t\in\Psi_{T_k+1}^s}\mathbb{E}\Big[\mathsf{Rev}_t\big(\tilde{p}_t^*\big)-\mathsf{Rev}_t\big(p_t\big)\Big]\\
&\le \sum_{s=1}^{S}\Big(8B\cdot 2^{-s}+(4s-2)BL\mathbb{E}_x\left[|\hat{v}_k(x)-v^*(x)|\right]\Big)|\Psi_{T_k+1}^s|.
\end{aligned}$$

where the last inequality is due to Lemma 4.3.

From Lemma C.3, we have

$$|\Psi_{T_k+1}^s|\le 2^s\sqrt{2N_k|\Psi_{T_k+1}^s|\ln(T_kN_k/\delta)},$$

which implies

$$\begin{aligned}
\sum_{s=2}^{S}8B\cdot 2^{-s}|\Psi_{T_k+1}^s| &\le \sum_{s=2}^{S}8B\sqrt{2N_k|\Psi_{T_k+1}^s|\ln(T_kN_k/\delta)}\\
&\le 8B\sqrt{2N_k(S-1)\sum_{s=2}^{S}|\Psi_{T_k+1}^s|\ln(T_kN_k/\delta)}\\
&\le 8B\sqrt{2N_kT_k\ln(T_kN_k/\delta)\ln T_k}.
\end{aligned}$$

Additionally, we have

$$\mathbb{E}\left[\sum_{t\in\Psi^1_{T_k+1}}\mathcal{R}^1_t\right]\le 2B|\Psi^1_{T_k+1}|\le 8N_k\ln(T_kN_k/\delta).$$

Putting above inequalities back yields

$$\begin{aligned}&\sum_{s=2}^{S}8B\cdot 2^{-s}|\Psi^s_{T_k+1}|+(4s-2)BL\mathbb{E}_x[|\hat{v}_k(x)-v^*(x)|]|\Psi^s_{T_k+1}|\\ \le&8B\sqrt{2N_kT_k\ln(T_kN_k/\delta)\ln T_k}+BL\mathbb{E}_x[|\hat{v}_k(x)-v^*(x)|]\sum_{s=2}^{S}(4s-2)|\Psi^s_{T_k+1}|\\ \le&8B\sqrt{2N_kT_k\ln(T_kN_k/\delta)\ln T_k}+BL\mathbb{E}_x[|\hat{v}_k(x)-v^*(x)|]T_k(4\ln T_k-2).\end{aligned}$$

The second inequality is the Cauchy-Schwarz inequality.

Notice that $\log_2\sqrt{T_k}\le\log_2\sqrt{\ell_k}=\frac{k}{3}$ and

$$\mathbb{E}_x[|\hat{v}_k(x)-v^*(x)|]\le\sqrt{\mathbb{E}_x[(\hat{v}_k(x)-v^*(x))^2]}\le\ln^{\frac{1}{3}}(k|\mathcal{V}|/\delta)(k\ell_k)^{-\frac{1}{3}}.$$

due to Lemma 3.1. We obtain the desired result by summing over all episodes. □

Lemma 4.5. Under Assumption 2.1, the expected discretization regret $\mathbb{E}\left[\sum_{t=1}^{T}\mathcal{R}^2_t\right]$ is upper bounded by

$$\mathbb{E}\left[\sum_{t=1}^{T}\mathcal{R}^2_t\right]\le\sum_{k=1}^{\lceil\log_2 T\rceil}\frac{4BT_k}{N_k}.$$

Proof. We consider the learning regret at each round t in the episode k. It admits the form

$$p^*_t(1-F(p^*_t-v^*(x_t)))-\tilde{p}^*_t(1-F(\tilde{p}^*_t-v^*(x_t))),$$

where p^*_t is the overall best price and $\tilde{p}^*_t$ is the discrete best price among the candidate set.

Denote $\dot{p}_t=\max\{0,\max_{j\in[N_k],m_j+\hat{v}_k(x_t)\le p^*_t}(m_j+\hat{v}_k(x_t))\}$. Then we discuss case by case.

1. $\{j\in[N_k],m_j+\hat{v}_k(x_t)\le p^*_t\}=\emptyset$. Then we have $\dot{p}_t=0$ and

$$p^*_t\le m_1+\hat{v}_k(x_t)\le-\|\hat{v}_k\|_\infty+\frac{B+2\|\hat{v}_k\|_\infty}{2N_k}+\|\hat{v}_k\|_\infty\le\frac{B+2\|\hat{v}_k\|_\infty}{2N_k}\le\frac{3B}{2N_k}.$$

Therefore, we have $0\le p^*_t-\dot{p}_t\le\frac{3B}{2N_k}$.

2. $\{j\in[N_k],m_j+\hat{v}_k(x_t)\le p^*_t\}\ne\emptyset$ and $\max_{j\in[N_k],m_j+\hat{v}_k(x_t)\le p^*_t}(m_j+\hat{v}_k(x_t))<m_{N_k}+\hat{v}_k(x_t)$. Then we have $\max_{j\in[N_k],m_j+\hat{v}_k(x_t)\le p^*_t}(m_j+\hat{v}_k(x_t))\le\dot{p}_t\le\max_{j\in[N_k],m_j+\hat{v}_k(x_t)\le p^*_t}(m_j+\hat{v}_k(x_t))+\frac{3B}{2N_k}$. Namely, we have $0\le p^*_t-\dot{p}_t\le\frac{3B}{2N_k}$.

3. $\{j \in [N_k], m_j + \hat{v}_k(x_t) \le p_t^*\} \neq \emptyset$ and $\max_{j\in[N_k], m_j+\hat{v}_k(x_t)\le p_t^*}(m_j + \hat{v}_k(x_t)) = m_{N_k} + \hat{v}_k(x_t)$. Since $m_{N_k} + \hat{v}_k(x_t) \ge B + \hat{v}_k(x_t) - \frac{3B}{2N_k} - \|\hat{v}_k\|_\infty \ge B - \frac{3B}{2N_k}$, then we have $B - \frac{3B}{2N_k} \le m_{N_k} + \hat{v}_k(x_t) \le p_t^* \le B$. Therefore we have $p_t^* - \dot{p}_t \le \frac{3B}{2N_k}$.

From above discussion, we conclude that $0 \le p_t^* - \dot{p}_t \le \frac{3B}{2N_k}$. Hence we have

$$\begin{aligned}
&p_t^*(1 - F(p_t^* - v^*(x_t))) - \dot{p}_t^*(1 - F(\dot{p}_t^* - v^*(x_t))) \\
\le& p_t^*(1 - F(\dot{p}_t - v^*(x_t))) - \dot{p}_t^*(1 - F(\dot{p}_t^* - v^*(x_t))) \\
\le& (p_t^* - \dot{p}_t)(1 - F(\dot{p}_t - v^*(x_t))) \\
\le& \frac{3B}{2N_k}.
\end{aligned}$$

We first sum over all rounds in the episode k and then sum over all episodes. We can obtain the desired results. □

Theorem 4.1. Under Assumption 2.1 and Assumption 2.2, the expected regret satisfies

$$\mathbb{E}[\mathrm{Reg}(T)] = \mathcal{O}\left(BT^{\frac{2}{3}} \ln^{\frac{1}{3}}\left(\delta^{-1}|\mathcal{V}| \ln T\right) \ln^{\frac{2}{3}}\left(\delta^{-1} T \ln T\right)\right)$$

with probability at least $1 - \delta$.

Proof. Since the price p_t is bounded by B, then the regret caused by the exploration phase is

$$\sum_{k=1}^{\lceil \log_2 T \rceil} BT_k^e \le B \sum_{k=1}^{\lceil \log_2 T \rceil} (k\ell_k)^{\frac{2}{3}} \ln^{\frac{1}{3}}(|\mathcal{V}|/\delta) \le 2BT^{\frac{2}{3}} \ln^{\frac{1}{3}}(|\mathcal{V}|/\delta) \log_2^{\frac{2}{3}} T.$$

From the condition $T_k^e > \ell_k$, we know that the UCB phase will starts at least after $k^* = \lceil \log_2(\ln(\delta^{-1}|\mathcal{V}| \log_2 T)) \rceil$. The regret caused by the exploration phase up to k^* is at most $B2^{k^*} \le B \ln(\delta^{-1}|\mathcal{V}| \log_2 T)$ regret.

We then combine the results in Lemma 4.4 and Lemma 4.5 and obtain

$$\begin{aligned}
\mathbb{E}[\mathrm{Reg}(T)] &= \mathbb{E}\left[\sum_{t=1}^{T} \mathcal{R}_t^1\right] + \mathbb{E}\left[\sum_{t=1}^{T} \mathcal{R}_t^2\right] \\
&\le \sum_{k=1}^{\lceil \log_2 T \rceil} \Big(8B\sqrt{2N_k T_k \ln(T_k N_k/\delta) \ln T_k} + \frac{3BT_k}{2N_k} + 4BL \ln^{\frac{1}{3}}(|\mathcal{V}| \ln T/\delta)(k\ell_k)^{\frac{2}{3}} \\
&\qquad + 2B\sqrt{T_k} + 8N_k \ln(T_k N_k/\delta)\Big).
\end{aligned}$$

Since $N_k = \lceil T_k^{\frac{1}{3}} / \ln^{\frac{2}{3}}(T_k/\delta) \rceil$, we have

$$\begin{aligned}
\mathbb{E}[\mathrm{Reg}(T)] \le\ & 15BT^{\frac{2}{3}} \ln^{\frac{2}{3}}(T \log_2(T)/\delta) + 8BLT^{\frac{2}{3}} \ln^{\frac{1}{3}}(|\mathcal{V}| \log_2 T/\delta) \log_2^{\frac{2}{3}} T \\
&+ B \ln(\delta^{-1}|\mathcal{V}| \log_2 T) + B\sqrt{T}(1 + \log_2 T) + 8T^{\frac{1}{3}} \ln^{\frac{1}{3}}(T/\delta).
\end{aligned}$$

□

C.3 Proofs for lower bounds in Section 4.2

Construction of noise distribution. We start with an infinitely differentiable function defined as

$$u_0(x) = \begin{cases} \exp\left(-\frac{1}{x\left(\frac{1}{3}-x\right)}\right) & \text{if } x \in \left(0, \frac{1}{3}\right), \\ 0 & \text{otherwise.} \end{cases}$$

It ia easy to see $u_0(x)$ is non-negative. We then normalize u_0 and define the following function

$$u(x) = \left(\int_0^{\frac{1}{3}} u_0(t)dt\right)^{-1} \int_{-\infty}^{x} u_0(t)dt.$$

Notice that $u(x) = 0$ for $x \le 0$ and $u(x) = 1$ for $x \ge \frac{1}{3}$. For any positive integer $l \in \mathbb{N}$, we consider the l-th order derivative of $u(x)$ on the interval $(0, \frac{1}{3})$:

$$\frac{\text{poly}(x)}{(x(\frac{1}{3}-x))^{2l-2}} \exp\left(-\frac{1}{x(\frac{1}{3}-x)}\right),$$

This expression suggests that $u^{(m)}(x)$ is upper bounded by a constant L_m over $\mathbb{R}$. We introduce the subscript m to indicate that the constant depends on m. Moreover, $u^{(m)}(x)$ satisfies $u^{(m)}(0) = 0$ and $u^{(m)}(\frac{1}{3}) = 0$ for $m \ge 1$. Therefore, if we consider the Taylor expansion of $u(x)$ at $x = \frac{1}{3}$ with Lagrange remainder, we know that $u(x) = u(\frac{1}{3}) + \frac{(x-\frac{1}{3})^m}{m!} u^{(m)}(\xi)$ for $\xi \in (\min(\frac{1}{3}, x), \max(\frac{1}{3}, x))$. Since $u^{(m)}(x)$ is bounded, we conclude that $|u(\frac{1}{3}) - u(x)| \le L_m |x - \frac{1}{3}|^m$ for $L_m = \max_x u^{(m)}(x)/m!$.

We summarize the properties of $u(x)$ in the following proposition.

Proposition C.1. 1. $u(x)$ is non-decreasing.

2. For any positive integer m, $u^{(m)}(0) = 0$ and $u^{(m)}(\frac{1}{3}) = 0$.
3. For any positive integer m and any $x \in \mathbb{R}$, $|u(\frac{1}{3}) - u(x)| \le L_m |x - \frac{1}{3}|^m$ for $L_m > 0$.

Next step is to construct a bump function

$$B(x) = \begin{cases} 0 & \text{if } x < 0, \\ u(x) & \text{if } 0 \le x \le \frac{1}{3}, \\ 1 & \text{if } \frac{1}{3} < x < \frac{2}{3}, \\ u(1-x) & \text{if } \frac{2}{3} \le x \le 1, \\ 0 & \text{if } x > 1. \end{cases}$$

Since $u(x)$ is infinitely differentiable and its m-th order derivatives vanish at 0 and $\frac{1}{3}$, $B(x)$ is also infinitely differentiable. Furthermore, we have

$$\left|B(x) - B\left(\frac{1}{3}\right)\right| \le L_m \left|x - \frac{1}{3}\right|^m \tag{EC.1}$$

from Proposition C.1. We also define the rescaled bump function for any $a < b$ as

$$B_{[a,b]}(x) = B\left(\frac{x-a}{b-a}\right).$$

The above inequality (EC.1) translates to

$$B_{[a,b]}\left(a+\frac{b-a}{3}\right)-B_{[a,b]}(x)=B\left(\frac{1}{3}\right)-B\left(\frac{x-a}{b-a}\right)\le L_m\left|\frac{1}{3}-\frac{x-a}{b-a}\right|^m=\frac{L_m}{(b-a)^m}\left|a+\frac{b-a}{3}-x\right|^m.$$

We proceed by defining a series of intervals within the range $[0,1]$, denoted as $[0,1]=[a_0,b_0]\supset[a_1,b_1]\supset\cdots\supset[a_k,b_k]\supset\cdots$. The lengths of these intervals follow the pattern $w_k=b_k-a_k=3^{-k!}$ for $k\ge 1$, while $w_0=b_0-a_0=1$. To select each interval $[a_k,b_k]$, we begin by dividing the range $[a_{k-1}+\frac{w_{k-1}}{3},b_{k-1}-\frac{w_{k-1}}{3}]$ into $Q_k=\frac{w_{k-1}}{3w_k}$ sub-intervals of equal length w_k. We then randomly choose one of these sub-intervals as $[a_k,b_k]$. From construction, we know that $B_{[a_k,b_k]}(x)$ is a constant in $[a_k+\frac{w_k}{3},b_k-\frac{w_k}{3}]$. Therefore, we have $B^{(l)}_{[a_k,b_k]}(x)=0$ on any $x\in[a_k+\frac{w_k}{3},b_k-\frac{w_k}{3}]$ for arbitrary positive integer l. Importantly, there exist infinitely many series of intervals that can be constructed in this manner. For each of these interval series, we are able to define the function

$$f(x)=c_f\sum_{k=0}^{\infty}w_k^m B_{[a_k,b_k]}(x),$$

where c_f is a constant remained to be determined later. There are a few properties of $f(x)$ shown in the following proposition.

Proposition C.2. 1. $f(x)\in[0,\frac{3}{2}c_f]$ for any $x\in\mathbb{R}$.

2. There exists a unique $x^*\in[0,1]$ such that $f(x^*)=\max_{x\in[0,1]}f(x)=c_f\sum_{k=0}^{\infty}w_k^m$. Specifically, $\{x^*\}=\bigcap_{k=1}^{\infty}[a_k,b_k]$.

3. $f(x)$ is unimodal around the unique maximizer x^*.

4. $f(x)$ is m-th differentiable and $|f^{(m)}(x)|\le c_f L_m$.

Proof.

1. Since $0\le B_{[a_k,b_k]}(x)\le 1$, we have

$$0\le f(x)\le\sum_{k=0}^{\infty}c_f w_k^m\le c_f\sum_{k=0}^{\infty}3^{-k}\le\frac{3}{2}c_f<\infty.$$

2. Since $w_k=b_k-a_k=3^{-k!}\to 0$, then there exists a unique x^* such that $x^*\in[a_k,b_k]$ for any k. For $B_{[a_k,b_k]}(x)$, it is non-decreasing on $(-\infty,x^*]$ and non-increasing on $[x^*,+\infty)$. Hence, x^* is the maximizer of $f(x)$. Since $x^*\in[a_{k+1},b_{k+1}]$ and thus $B_{[a_k,b_k]}(x)=1$, we immediately know that the maximum is $c_f\sum_{k=0}^{\infty}w_k^m\le\frac{3}{2}c_f$.

3. For any $k\ge 0$, $B_{[a_k,b_k]}(x)$ is non-decreasing in $[0,x^*]$ and non-increasing in $[x^*,1]$ since $x^*\in[a_{k+1},b_{k+1}]\subset[a_k,b_k]$. Thus $f(x)$, as the sum of these bump functions, is non-decreasing in $[0,x^*]$ and non-increasing in $[x^*,1]$.

4. Since $B(x)$ is infinitely times differentiable, we have

$$\sum_{k=0}^{\infty} w_k^m B^{(m-1)}(\frac{x-a_k}{b_k-a_k})(b_k-a_k)^{m-1} \leq \sum_{k=0}^{\infty} w_k \leq \sum_{k=0}^{\infty} 3^{-k} < \infty,$$

which shows that $c_f \sum_{k=0}^{\infty} w_k^m B_{[a_k,b_k]}^{(m-1)}(x)$ converges absolutely and uniformly. The above result implies $f(x)$ is $(m-1)$-th differentiable.

Notice that for any $x \in [0,1]$, there exists at most one k, such that $B_{[a_k,b_k]}^{(m)}(x) \neq 0$. Consider the case when $B_{[a_k,b_k]}^{(m)}(x) \neq 0$, and we know that

$$x \in [a_k, b_k] \subset [a_{k-1} + \frac{w_{k-1}}{3}, b_{k-1} - \frac{w_{k-1}}{3}] \subset \cdots [a_0, b_0] = [\frac{1}{3}, \frac{2}{3}].$$

Hence, we know that $B_{[a_j,b_j]}^{(m)}(x) = 0$ for $j = 0, 1, \cdots, k-1$. Moreover, we know that $x \notin [a_{k+1}, b_{k+1}]$; otherwise we have $B_{[a_k,b_k]}^{(m)}(x) = 0$. Therefore, we have $x \notin [a_j, b_j]$ for $j = k+1, k+2, \cdots$, which indicates $B_{[a_j,b_j]}^{(m)}(x) = 0$. Therefore, we have

$$c_f \max_{x \in [a_k,b_k], k \in \mathbb{N}} |w_k^m B_{[a_k,b_k]}^{(m)}(x)| \leq c_f \max_{x \in [a_k,b_k], k \in \mathbb{N}} |B^{(m)}(\frac{x-a_k}{b_k-a_k})| \leq c_f m! L_m.$$

The above property implies $c_f \sum_{k=0}^{\infty} w_k^m B_{[a_k,b_k]}^{(m)}(x)$ also converges absolutely and uniformly. From the above property, we know that f is m-th differentiable.

□

We then rescale $f(x)$ to $[0,1]$ and define the function $g(x) = 1 - \frac{1}{1+f(x)}$. Then we have

$$|g(x^*) - g(x)| = \left| \frac{1}{1+f(x^*)} - \frac{1}{1+f(x)} \right| \leq |f(x^*) - f(x)| \leq \tilde{L}_m |x^* - x|^m.$$

In addition, $g(x)$ is unimodal with the same unique maximizer x^* of $f(x)$.

We further define the function $F(x)$ as

$$F(x) = \begin{cases} 0, & \text{if } x < b, \\ 1 - \frac{b}{x} - \frac{1-b}{x} g\left(\frac{x-b}{1-b}\right), & \text{if } b \leq x \leq 1, \\ 2 - \frac{1+b}{x}, & \text{if } 1 < x \leq 1+b, \\ 1, & \text{if } x > 1+b. \end{cases}$$

We specify the choices of c_f and b. Let $c_f \in (0, \frac{1}{L_1})$ and $b = \frac{1+c_f L_1}{2} \in (0,1)$.

Proposition C.3. 1. $F(x)$ is a cumulative distribution function for some $\mathbb{R}$-valued random variable.

2. $F(x)$ is m-th differentiable on the interval $(b,1)$.

3. There exists a unique minimizer $x_r^* \in [b,1]$ for the revenue function $\mathsf{Rev}(x) = x(1-F(x))$.

Proof.

1. It is easy to check that $F(-\infty)=0, F(+\infty)=1$ and $F(x)$ is continuous on $\mathbb{R}$. The remaining step is to show that $F(x)$ is non-decreasing. The monotonicity of $F(x)$ is clear on $(-\infty,b)\cup(1,+\infty]$. Then we consider the derivative of $F(x)$ on $[b,1]$. We have

$$F'(x)=\frac{b-xg'(\frac{x-b}{1-b})+(1-b)g(\frac{x-b}{1-b})}{x^2}.$$

Since $|g'(x)|=\left|\frac{f'(x)}{(1+f(x))^2}\right|\le c_f L_1$, we have

$$b-xg'\left(\frac{x-b}{1-b}\right)\ge b-c_f L_1=\frac{1-c_f L_1}{2}>0.$$

It indicates that $F'(x)>0$ on $[b,1]$. Note that $F(b)\le F(x)\le F(1)$ for $x\in[b,1]$. Thus $F(x)$ is non-decreasing on $\mathbb{R}$.

2. It follows directly that $f(x)$ is m-th differentiable.

3. Simple calculation yields

$$\mathsf{Rev}(x)=\begin{cases}x, & \text{if } x\in[0,b),\\ b+(1-b)g\left(\frac{x-b}{1-b}\right), & \text{if } x\in[b,1],\\ 1+b-x, & \text{if } x\in(1,1+b],\\ 0, & \text{if } x\in(1+b,\infty).\end{cases}$$

It is easy to check $\mathsf{Rev}(x)\ge b>rev(y)$ for any $x\in[b,1]$ and $y\in[0,+\infty)-[b,1]$. Since $g(x)$ has the same unique maximizer $x_g^*=x_f^*$ for f, we obtain that

$$x_r^*=b+(1-b)x_g^*\in[b,1]$$

is the unique maximizer for $\mathsf{Rev}(x)$.

□

Note that we can set v^* as a constant as that the zero-mean requirement of F can be relaxed.

Construction of instances. Now, we proceed to construct instances to prove the main theorem. It is important to note that there is a bijection between each interval series $[a_k,b_k]_{k\ge0}$ and a triplet $(f(v),g(v),F(v))$. Therefore, each interval series corresponds to a noise distribution $F(x)$ and, consequently, a problem instance constructed as described above. With well-formulated groups of such instances, we will prove that no policy can perform well on all the instances in one group.

For each k, let $n_k=\lceil\frac{w_{k-1}}{w_k^m}\rceil$. Then we consider the possible Q_k choices of $[a_k,b_k]$. Denote these intervals as $I_j, j=1,2,\cdots,Q_k$ and the corresponding triplet $(f(x),g(x),F(x))$ for $[a_k,b_k]=I_j$ as $(f_j(x),g_j(x),F_j(x))$. These noise CDF $\{F_j(x)\}_{j\in[Q_k]}$ form the k-th group of instances. We further define a reference function $f_0(x)=c_f\sum_{j=0}^{k-1}w_j^m B_{[a_j,b_j]}(x)$, which corresponds to the finite series of

intervals $\{[a_j, b_j]\}_{0\le j\le k-1}$. We can define $g_0(x)$ and $F_0(x)$ based on $f_0(x)$ through the previously introduced procedure.

Consider an arbitrary policy π. This policy interacts with the noise distribution $F_j(x)$ and generates a sequence of prices and responses, denoted as $\mathcal{H}_n = (p_1, y_1, p_2, y_2, \cdots, p_n, y_n)$. Denote the distribution of u_n under $F_j(x)$ as $\mathbb{P}_j$ for $j \in \{0, 1, \cdots, Q_k\}$. It is reasonable to assume that $p_t \in [b, 1]$, as any price $p' \in [0, B] - [b, 1]$ would be suboptimal and dominated by any price $p \in [b, 1]$ due to the properties of the revenue function.

Now we derive the KL-divergence $D_{\mathrm{KL}}(\mathbb{P}_0||\mathbb{P}_j)$ for any $j \in [Q_k]$. Since given the price p, the response y follows the Bernoulli distribution with parameter $1 - F_j(p)$. Then we have

$$\begin{aligned}
D_{\mathrm{KL}}(\mathbb{P}_0||\mathbb{P}_j) &= \sum_{t=1}^{n_k} \mathbb{E}_{\mathbb{P}_0}\left[D_{\mathrm{KL}}(Ber(1-F_0(p_t))||Ber(1-F_j(p_t)))\right] \\
&= \sum_{t=1}^{n_k} \mathbb{E}_{\mathbb{P}_0}\Bigg[D_{\mathrm{KL}}(\mathbf{I}_{\{\frac{p_t-b}{1-b}\in I_j\}} Ber(1-F_0(p_t))||Ber(1-F_j(p_t))) + \\
&\qquad D_{\mathrm{KL}}(\mathbf{I}_{\{\frac{p_t-b}{1-b}\notin I_j\}} Ber(1-F_0(p_t))||Ber(1-F_j(p_t)))\Bigg]
\end{aligned}$$

The following lemma will help us to show $D_{\mathrm{KL}}(1-F_0(p_t)||1-F_j(p_t)) = O(w_k^{2m})$.

Lemma C.4. (Xu and Wang 2022) For Bernoulli distributions $Ber(p)$ and $Ber(p+\epsilon)$ with $\frac{1}{2} \le p \le p+\epsilon \le \frac{1}{2} + C$, we have

$$D_{\mathrm{KL}}(Ber(p)||Ber(p+\epsilon)) \le \frac{4}{1-4C^2}\epsilon^2.$$

For simplicity, denote $q_t = \frac{p_t - b}{1-b}$. Then for p_t such that $q_t \in I_j$, we have $D_{\mathrm{KL}}(1-F_0(p_t)||1-F_j(p_t)) = 0$ by their constructions. Then we focus on the p_t such that $q_t \in I_j$, we have

$$\begin{aligned}
(1-F_j(p_t)) - (1-F_0(p_t)) &= \frac{1-b}{p_t}(g_j(q_t) - g_0(q_t)) \\
&= \frac{1-b}{p_t}\left(\frac{1}{1+f_0(q_t)} - \frac{1}{1+f_j(q_t)}\right) \\
&= \frac{1-b}{p_t}\left(\frac{c_f \sum_{i=k}^{\infty} w_i^m B_{[a_i,b_i]}(q_t)}{(1+f_0(q_t))(1+f_j(q_t))}\right) \\
&\le \frac{1-b}{b}\left(c_f w_i^m \sum_{i=k}^{\infty} 3^{-i}\right) \\
&\le \frac{3}{2} c_f w_k^m
\end{aligned}$$

From the third equality, we also know that $(1-F_j(p_t)) - (1-F_0(p_t)) \ge 0$. Moreover, we have

$$1 - F_0(p_t) = \frac{b + (1-b)g_0(q_t)}{p_t} \ge b \ge \frac{1}{2}$$

since $g(q_t)>0$. Moreover, we have $q_t=\frac{p_t-b}{1-b}\in I_j\subset[\frac{1}{3},\frac{2}{3}]$ and thus $p_t\geq b+\frac{1-b}{3}$. Therefore, we obtain

$$1-F_j(p_t)\leq 1-F_j(b+\frac{1-b}{3})=\frac{3b+3(1-b)\frac{f_j(1/3)}{f_j(1/3)+1}}{2b+1}=\frac{3b+3(1-b)\frac{c_f}{c_f+1}}{2b+1}=\frac{3(b+c_f)}{(2b+1)(c_f+1)},$$

where the second equality is from $f_j(\frac{1}{3})=c_f$. We further choose $c_f<\frac{1}{L_1+6}$ and recall that $b=\frac{1+c_fL_1}{2}$. Hence we have $3(b+c_f)<2b+1$ and obtain that

$$1-F_j(p_t)\leq\frac{1}{1+c_f}<1.$$

Then by Lemma C.4, we obtain

$$\begin{aligned}D_{\mathrm{KL}}(1-F_0(p_t)||1-F_j(p_t))&\leq\frac{4}{1-4(\frac{1}{1+c_f}-\frac{1}{2})^2}((1-F_j(p_t))-(1-F_0(p_t)))^2\\&=\frac{9}{4}c_f(1+c_f)^2w_k^{2m}\\&\leq\frac{1}{300}w_k^{2m}.\end{aligned}$$

The last inequality can hold as we can choose positive c_f such that $0<c_f<\min\{10^{-4},\frac{1}{L_1+6}\}$.

We can further bound $D_{\mathrm{KL}}(\mathbb{P}_0||\mathbb{P}_j)$ as

$$\begin{aligned}D_{\mathrm{KL}}(\mathbb{P}_0||\mathbb{P}_j)&=\sum_{t=1}^{n_k}\mathbb{E}_{\mathbb{P}_0}\Big[D_{\mathrm{KL}}(\mathbf{I}_{\{\frac{p_t-b}{1-b}\in I_j\}}Ber(1-F_0(p_t))||Ber(1-F_j(p_t)))+\\&\qquad D_{\mathrm{KL}}(\mathbf{I}_{\{\frac{p_t-b}{1-b}\notin I_j\}}Ber(1-F_0(p_t))||Ber(1-F_j(p_t)))\Big]\\&\leq\sum_{t=1}^{n_k}\mathbb{E}_{\mathbb{P}_0}\left[\mathbf{I}_{\{\frac{p_t-b}{1-b}\notin I_j\}}\times 0+\mathbf{I}_{\{\frac{p_t-b}{1-b}\in I_j\}}\times\frac{1}{300}w_k^{2m}\right]\\&=\frac{1}{300}w_k^{2m}\sum_{t=1}^{n_k}\mathbb{E}_{\mathbb{P}_0}\left[\mathbf{I}_{\{\frac{p_t-b}{1-b}\in I_j\}}\right]\end{aligned}$$

To bound the last term, we need to following lemma.

Lemma C.5. (Luo et al. 2024) Consider any function h on the sequence u that has a bounded value range $[0,M]$. Then for two probability measure $\mathbb{P}_0$ and $\mathbb{P}_j$, we have

$$\mathbb{E}_{\mathbb{P}_j}[h(u)]-\mathbb{E}_{\mathbb{P}_0}[h(u)]\leq M\sqrt{\frac{1}{2}D_{\mathrm{KL}}(\mathbb{P}_j||\mathbb{P}_0)}.$$

For notation brevity, denote $N_j=|\{t\in[n_k]|q_t\in I_j\}|$ for each $j\in[Q_k]$. Notice that N_j is a function of the price and response sequence u_n and is bounded in the range $[0,n_k]$. Thus combing above inequalities, we obtain

$$\begin{aligned}\mathbb{E}_{\mathbb{P}_j}[N_j]-\mathbb{E}_{\mathbb{P}_0}[N_j]&\leq n_k\sqrt{\frac{1}{2}D_{\mathrm{KL}}(\mathbb{P}_j||\mathbb{P}_0)}\\&\leq n_k\sqrt{\frac{1}{600}w_k^{2m}\mathbb{E}_{\mathbb{P}_0}[N_j]}\end{aligned}$$

For $k \geq 3$, we sum over $j \in [Q_k]$ take the average to obtain

$$\begin{aligned}
\frac{1}{Q_k}\sum_{j=1}^{Q_k}\mathbb{E}_{\mathbb{P}_j}[N_j] &\leq \frac{1}{Q_k}\sum_{j=1}^{Q_k}\mathbb{E}_{\mathbb{P}_0}[N_j] + \frac{n_k w_k^m}{20Q_k}\sum_{j=1}^{Q_k}\sqrt{\mathbb{E}_{\mathbb{P}_0}[N_j]} \\
&= \frac{n_k}{Q_k} + \frac{n_k w_k^m}{20Q_k}\sum_{j=1}^{Q_k}\sqrt{\mathbb{E}_{\mathbb{P}_0}[N_j]} \\
&\leq \frac{n_k}{Q_k} + \frac{n_k w_k^m}{20Q_k}\sqrt{Q_k\sum_{j=1}^{Q_k}\mathbb{E}_{\mathbb{P}_0}[N_j]} \\
&= \frac{n_k}{Q_k} + \frac{n_k w_k^m}{20Q_k}\sqrt{Q_k n_k} \\
&= n_k\left(\frac{1}{Q_k} + \frac{w_k^m}{20}\sqrt{\frac{2w_{k-1}}{w_k^{2m+1}Q_k}}\right) \\
&\leq n_k\left(\frac{1}{27} + \frac{\sqrt{6}}{20}\right) \\
&\leq \frac{1}{5}n_k
\end{aligned}$$

Therefore, there exists some $j \in [Q_k]$ such that $\mathbb{E}_{\mathbb{P}_j}[N_j] \leq \frac{n_k}{5}$. Denote the optimal price of the corresponding revenue function as p_j^*. Then $q_j^* = \frac{p_j^* - b}{1-b} \in I_j$ is the unique maximum for $f_j(x)$. For any $p \in [b,1]$ such that $q = \frac{p-b}{1-b} \notin I_j$, $B_{[a_i,b_i]}(q)$ is equal to zero for $i \geq k$ as $I_j \supset [a_{k+1}, b_{k+1}] \supset \cdots$. Recall that $f_j(x) = c_f\sum_{i=0}^{k-1} w_i^{2m}B_{[a_i,b_i]}(x) + c_f w_i^{2m}B_{[a_k,b_k]}(x) + c_f\sum_{i=k+1}^{\infty} w_i^{2m}B_{[a_i,b_i]}(x)$ Hence we have the following inequality:

$$f_j(q_j^*) - f_j(q) \geq c_f w_k^m.$$

We then have

$$g_j(q_j^*) - g_j(q) = \frac{f(q_j^*) - f(q)}{(1+f_j(q_j^*))(1+f_j(q))} \geq \frac{c_f}{(1+\frac{3}{2}c_f)^2}w_k^m$$

Therefore, we have

$$\mathsf{Rev}_j(p_j^*) - \mathsf{Rev}_j(p) = (1-b)(g_j(q_j^*) - g_j(q)) \geq \frac{1-c_f L_1}{2}\frac{c_f}{(1+\frac{3}{2}c_f)^2}w_k^m$$

For notation brevity, denote $\frac{1-c_f L_1}{2}\frac{c_f}{(1+\frac{3}{2}c_f)^2}$ as a positive constant $\tilde{C} > 0$.

Finally, we can compute the regret

$$\begin{aligned}
\mathbb{E}[\mathrm{Reg}(T)] &= \mathbb{E}_{\mathbb{P}_j}\left[\sum_{t=1}^{n_k}\Big(\mathsf{Rev}_j(p_j^*) - \mathsf{Rev}_j(p_t)\Big)\right] \\
&\geq \sum_{t=1}^{n_k}\mathbb{E}_{\mathbb{P}_j}\left[\mathbf{I}_{\{\frac{p_t-b}{1-b}\notin I_j\}}\Big(\mathsf{Rev}_j(p_j^*) - \mathsf{Rev}_j(p_t)\Big)\right] \\
&\geq \sum_{t=1}^{n_k}\mathbb{E}_{\mathbb{P}_j}\left[\mathbf{I}_{\{\frac{p_t-b}{1-b}\notin I_j\}}\tilde{C}w_k^m\right]
\end{aligned}$$

$$= \tilde{C} w_k^m \left(n_k - \mathbb{E}_{\mathbb{P}_j}[N_j] \right)$$
$$\geq 0.8 \tilde{C} w_k^m n_k.$$

Recall that $n_k = \lceil \frac{w_{k-1}}{k w_k^{2m+1}} \rceil$. We have

$$w_k^m n_k \geq n_k^{\frac{1+m-k^{-1}}{1+2m-k^{-1}}} \geq n_k^{\frac{1+m}{2m+1} - \frac{m}{k(2m+1)^2}}$$

Now we claim that for any $\delta > 0$, no policy can achieve the regret of $O(T^{\frac{m+1}{2m+1}-\delta})$. We prove by contradiction. Suppose that there is some $\delta > 0$ and a policy π such that $\mathrm{Reg}_\pi(T) \leq C_\pi T^{\frac{m+1}{2m+1}-\delta}$ for any noise distribution. There exists a sufficiently large k such that $\frac{m}{k(2m+1)^2} \leq \frac{\delta}{2}$ and $3^{-\frac{m \cdot (k-1)!}{2(2m+1)^2}} < \frac{0.8\tilde{C}}{C_\pi}$. Then we obtain

$$\mathrm{Reg}_\pi(n_k) \leq C_\pi n_k^{\frac{m+1}{2m+1}-\delta} < C_\pi n_k^{-\frac{m}{k(2m+1)^2}} n_k^{\frac{1+m}{2m+1} - \frac{m}{k(2m+1)^2}} < 0.8\tilde{C} n_k^{\frac{1+m}{2m+1} - \frac{m}{k(2m+1)^2}} < \mathrm{Reg}_\pi(n_k).$$

This forms a contradiction.

C.4 Proofs for Section 5

Assumption C.1. (Differentiablity) The noise distribution F is twice continuously differentiable.

Assumption C.2. (Concavity) The function F and $1 - F$ is log-concave.

Corollary 5.2. Assume that the noise distribution F is twice continuously differentiable, and F and $1 - F$ are log-concave. Under Assumption 5.1, the expected regret of Algorithm 3 satisfies

$$\mathbb{E}[\mathrm{Reg}(T)] = \mathcal{O}\left(\rho_{\mathcal{V}}(\delta) \ln T\right)$$

with probability at least $1 - \delta$.

Proof. Given the estimator $\hat{v}_k$ using data collected in the previous episode, p_t is only determined by x_t. Therefore, we know that the data collected in one episode is i.i.d. due to the i.i.d. nature of covariates. Thanks to Assumption C.1, we apply the Taylor's expansion at p_t^* to $\mathsf{Rev}_t(p_t)$:

$$\mathsf{Rev}_t(p_t^*) - \mathsf{Rev}_t(p_t) \leq C(p_t^* - p_t)^2,$$

where $C = \max_p |\mathsf{Rev}_t''(p)|$. Using first order conditions, for the optimal posted price p_t^*, we obtain

$$p_t^* = \frac{1 - F(p_t^* - v^*(x_t))}{F'(p_t^* - v^*(x_t))}.$$

Since $1 - F$ is log-concave, the hazard function $v \to \frac{F'(v)}{1-F(v)}$ is increasing, which implies $\phi(v) = v - \frac{1-F(v)}{F'(v)}$ is strictly increasing. Note that

$$\phi(p_t^* - v^*(x_t)) = p_t^* - v^*(x_t) - \frac{1 - F(p_t^* - v^*(x_t))}{F'(p_t^* - v^*(x_t))} = -v^*(x_t)$$

Define $g(v) = v + \phi^{-1}(-v)$ and we know that

$$g(v^*(x_t)) = v^*(x_t) + \phi^{-1}(-v^*(x_t)) = p_t^*.$$

Similarly, since p_t maximizes $p(1 - F(p - \hat{v}_k(x_t)))$, we know that $g(\hat{v}_k(x_t)) = p_t$. From lemma 15 in Javanmard and Nazerzadeh (2019), we know g is 1-Lipschitz and thus we have

$$|g(v^*(x_t)) - g(\hat{v}_k(x_t))| \le |v^*(x_t) - \hat{v}(x_t)|.$$

Therefore, we have

$$\mathsf{Rev}_t(p_t^*) - \mathsf{Rev}_t(p_t) \le C(p_t^* - p_t)^2 \le C|v^*(x_t) - \hat{v}(x_t)|^2.$$

Due to Assumption 5.1, we obtain

$$\mathbb{E}[\mathsf{Rev}_t(p_t^*) - \mathsf{Rev}_t(p_t)] \le \frac{C\rho_{\mathcal{V}}(\delta)}{2^{k-1}}.$$

Summing over all rounds in all episodes yields

$$\mathbb{E}[\mathrm{Reg}(T)] \le \sum_{k=1}^{\lceil \ln T \rceil} \sum_{t=2^{k-1}}^{2^k} \mathbb{E}[\mathsf{Rev}_t(p_t^*) - \mathsf{Rev}_t(p_t)] \le 3C\rho_{\mathcal{V}}(\delta) \ln T.$$

□

Corollary 5.3. Under Assumption 2.2 and Assumption 5.1, the expected regret of Algorithm 3 satisfies

$$\mathbb{E}[\mathrm{Reg}(T)] = \mathcal{O}\left(\sqrt{\rho_{\mathcal{V}}(\delta) T \ln T}\right)$$

with probability at least $1 - \delta$.

Proof. We first consider one-step regret:

$$\begin{aligned}
&\mathsf{Rev}_t(p_t^*) - \mathsf{Rev}_t(p_t) \\
=&p_t^*(1 - F(p_t^* - v^*(x_t))) - p_t(1 - F(p_t - v^*(x_t))) \\
=&p_t^*(1 - F(p_t^* - v^*(x_t))) - p_t^*(1 - F(p_t^* - \hat{v}_k(x_t))) + p_t^*(1 - F(p_t^* - \hat{v}_k(x_t))) - p_t(1 - F(p_t - \hat{v}_k(x_t))) \\
&\quad + p_t(1 - F(p_t - \hat{v}_k(x_t))) - p_t(1 - F(p_t - v^*(x_t))) \\
\le&p_t^*(1 - F(p_t^* - v^*(x_t))) - p_t^*(1 - F(p_t^* - \hat{v}_k(x_t))) + p_t(1 - F(p_t - \hat{v}_k(x_t))) - p_t(1 - F(p_t - v^*(x_t))) \\
\le&BL|v^*(x_t) - \hat{v}_k(x_t)|.
\end{aligned}$$

The first inequality is due to the optimality of p_t with respect to $\hat{v}_k$. Therefore, the Jensen's inequality yields

$$\mathbb{E}[\mathsf{Rev}_t(p_t^*) - \mathsf{Rev}_t(p_t)] \le BL\mathbb{E}[|v^*(x) - \hat{v}_k(x)|] \le BL\sqrt{\mathbb{E}[(v^*(x) - \hat{v}_k(x))^2|]}.$$

By summing over all rounds in all episodes and plugging in Assumption 5.1, we obtain the desired result. □